



Powering AI – Europe

In US footsteps, but very green. Well prepared.

- ◆ European data centre (DC) power demand follows in US footsteps and we expect it to quadruple in ten years
- ◆ European 1-3% annual demand growth for the next 25 years benefits renewables and electricity grids
- ◆ Europe is well prepared and welcomes growth/industrial opportunity. It's affordable and furthers independence

This report is the latest in our Powering AI series. Following our five-part series on the US, we turn our attention to Europe. We discuss core markets, growth in the context of total electricity demand, policy, affordability and implications for the sector.

European data centre power demand follows in US footsteps. European power demand from data centres more than quadruples from c90TWh in 2025 to >400TWh in 2035e on our numbers. By 2035, we expect European load to increase c27% compared to 2025; an annual growth rate of 1-3%. Circa 31% of this growth is from data centres.

European data centre growth broadens. We expect continued high data centre growth in tier 1 markets of Germany, the UK, France, the Netherlands and Ireland with expansion to and strong growth in new tier 2 markets in the Nordics, the South (Iberia & Italy) or the East, such as Poland. Expectations continue to increase with 2026 forecasts for data centre capacity by 2030 being 20% higher compared to 2025 estimates.

Homegrown electricity and DCs are an affordable strategic priority in Europe. As a fossil fuel importer at high gas prices, green homegrown electricity is an affordable strategic priority in Europe as are homegrown data centres. Affordability in Europe is not linked to data centres; it's rather a debate on taxation and lowering electricity prices relative to gas prices (to drive up electricity demand more). Given ambitions of electrification growth for over a decade, the system is well prepared.

Beneficiaries are electricity grids and renewables. With dual trends of data centres and energy independence, Europe's 4th era of power is a renewable (including offshore wind) and electric grid supercycle that benefits electric utilities, renewables developers and their supply chains. For ideas on how to play this theme, please see

[Utilities & Renewables: Capex supercycle improves](#), 11 August 2026

[Cable OEMs: Harnessing the power and fibre needs of data centres](#), 22 May 2026

[Assessing AI bottlenecks: Data centre equipment suppliers on the up](#), 25 March 2026.

This is our latest report on the Energy Transition theme. If you want to subscribe to any of our nine big themes, [click here](#).

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AI megadeals and the rest

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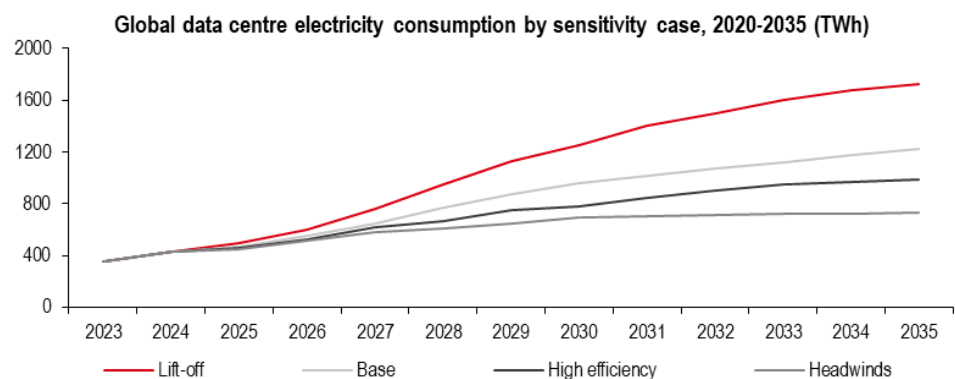
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Power demand and data centres

- ◆ European data centre (DC) power demand follows in US footsteps and we expect it to quadruple in ten years
- ◆ European power demand from data centres to rise from c90TWh in 2025 to >400TWh in 2035e
- ◆ In comparison, we expect US electricity demand from data centres to rise from c200TWh in 2025 to c900TWh in 2035e

Global data centre electricity consumption could triple or quadruple by 2035, according to the IEA's Stated Policies Scenario (STEPS)¹. This would contribute a moderate <10% to global electricity demand growth.

Global data centre electricity consumption could triple or quadruple by 2035



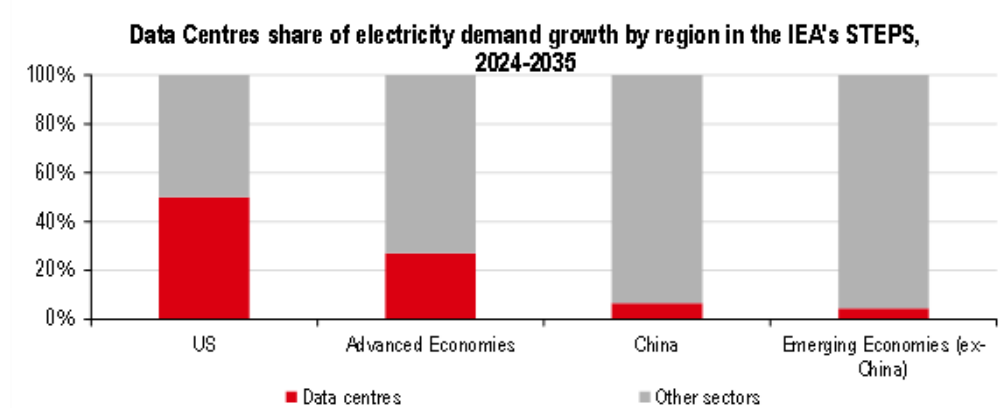
Source: IEA estimates

Global data centre growth will be highly concentrated, with the US dominating

Yet, data centre growth is expected to be highly concentrated, with 85% of new capacity additions over the next ten years in the US, Europe, and China. This is most stark in the US with data centres forecast to make up 50% of 2024-35 demand growth, compared to 6-10% in China.

¹ The International Energy Agency's (IEA) Stated Policies Scenario (STEPS) aims to project how the global energy sector might evolve based on actual policies already in place or officially announced by governments

Data centre growth will be highly concentrated

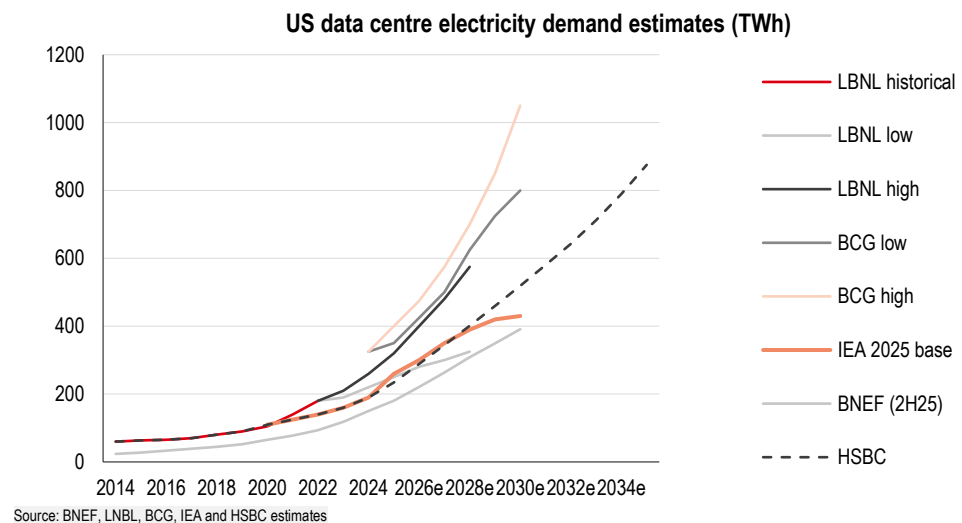


Source: IEA estimates

This note carries on from our five-part series on the US, where we expect load from data centres to more than quadruple

In the US, we expect electricity demand from data centres to rise from c200TWh in 2025 to c900TWh in 2035e and thus more than quadruple in ten years.

US data centre specific electricity demand forecasts include top-down chip delivery estimates as well as bottom-up project data bases

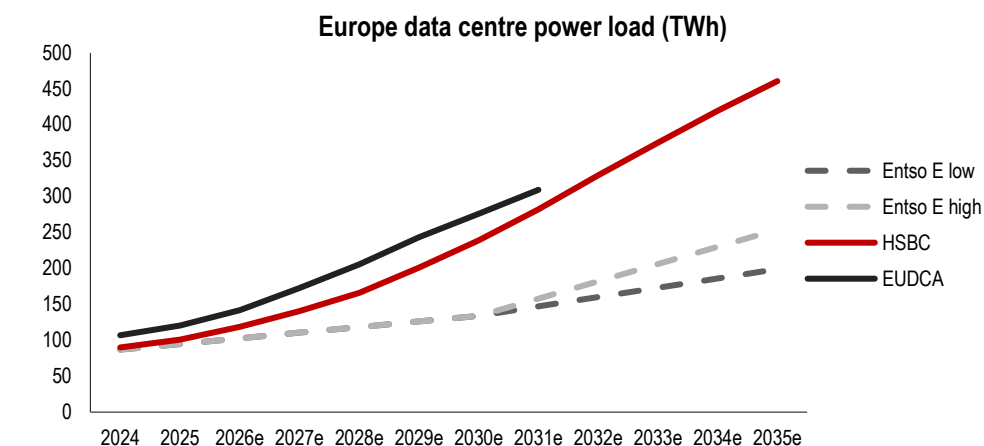


In our power AI – US series, we discussed data centre growth and implications for the US power system. In **part one**, we wrote about a [new era of load growth in the US](#) and the role of data centres. **Part two** covered [beneficiaries](#) in the power value chain while **part three** looked at [affordability](#) – are rising power prices linked to rising power demand and data centres? In **part four** we examined whether [grids are a bottle neck](#) while **Part five** highlighted sectors and [companies with exposure to the theme](#).

We expect European data centres demand to increase 4x in ten years

In Europe, we forecast electricity demand from data centres to rise from c90TWh in 2025 to >400TWh in 2035e and thus more than quadruple in ten years.

We forecast a strong rise in electricity demand from European data centres



Source: Entso-E forecasts, EUDCA (European Data Centre Association) IT capacity forecast converted to electricity load, HSBC estimates

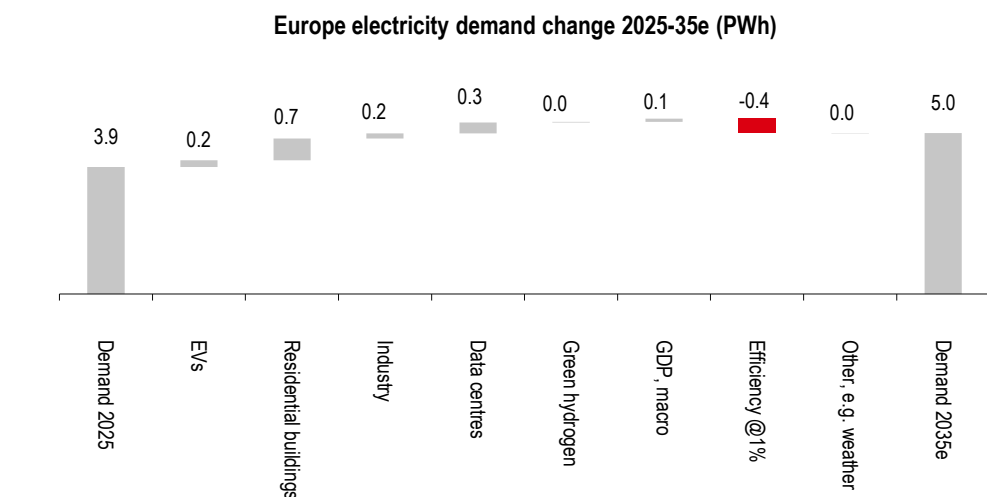
Therefore, in this note, we deep dive into European load growth expectations and the role that AI and data centres are playing in it.

Sustained growth for 25 years

- ◆ Europe on 1-3% annual demand growth for next 25 years
- ◆ The 4th era of power: renewables and electric grid investments supercycle from DC growth and energy independence goals
- ◆ US invest in gas; Europe invests more in grids and in offshore wind

In a 10-year view, we expect European load growth of 1.1PWh, a c27% increase compared to 2025. Circa 31% of growth expectations or 0.3PWh is driven by data centres with the remaining coming from EVs, building electrification, industry and weather offset by efficiencies.

In a 10-year view, we expect electricity demand growth of 1-3% per annum in Europe or 1.1PWh

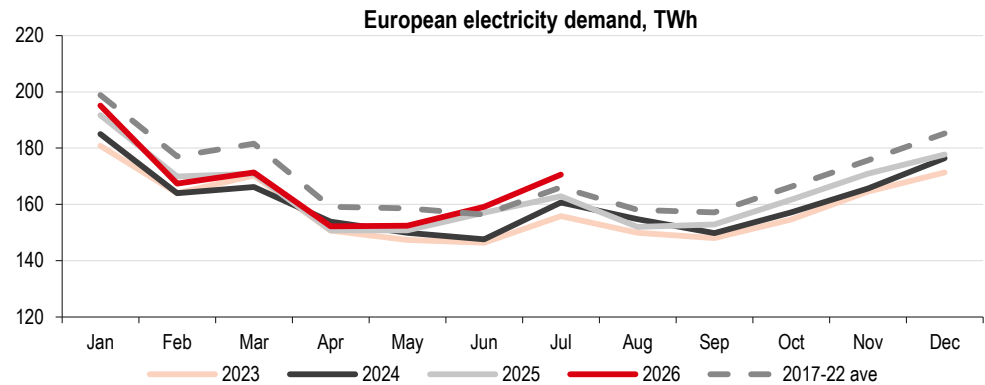


Source: HSBC estimates

Europe's electricity demand is showing signs of structural growth, but remain well below 2017-22 levels

European demand in 2026 remains well below 2017-22; the most meaningful trend in European power in the last five years was a c6% reduction during Europe's 2022/23 energy crisis. After moderate growth of 1-2% per annum in 2024 and 2025, 2026 demand is up 1.2% YoY, helped by a strong monthly surge of 5% YoY in July reflecting higher temperatures that boosted cooling demand. Will 2026 be affected again by efficiency effects (such as further farshoring and behaviour changes) as a reaction to the crisis in the Middle East?

European electricity demand remains well below 2017-22 levels, yet is starting to rebound and up 1.2% YTD

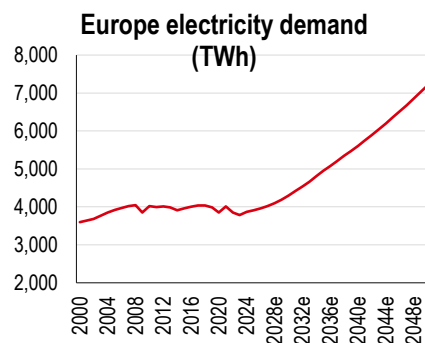


Source: Ember, HSBC

Electricity demand in Europe has been flat over the last 25 years

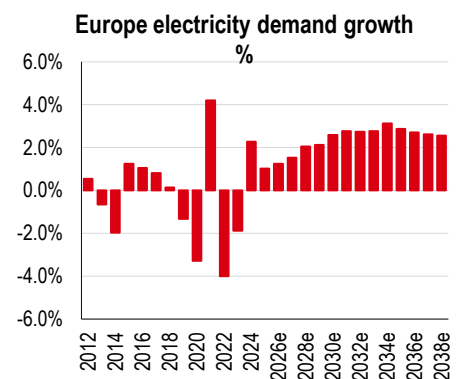
Over the last 25 years, electricity demand has been broadly flat in Europe, even slightly down in the last five years. For the next 25 years, we expect a new era of electricity demand growth at 1-3% per annum sustained by data centres and electrification drivers.

Over the last 25 years, electricity demand has been broadly flat in Europe



Source: Ember, HSBC estimates

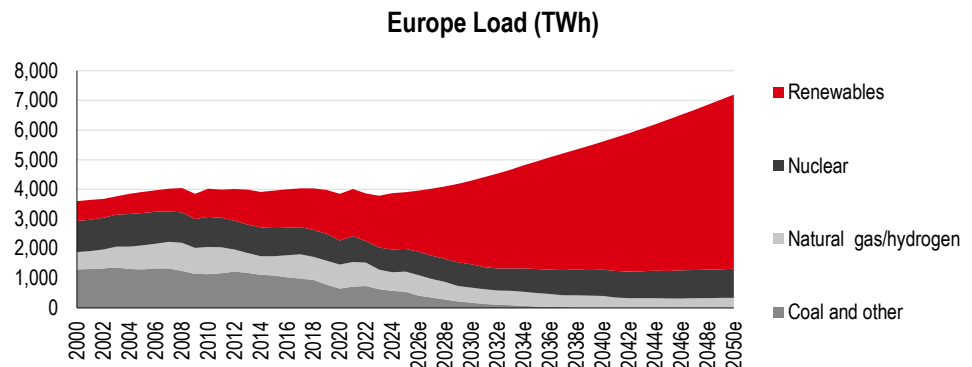
For the next 25 years, we expect a new era of electricity demand growth at 1-3% per annum



Source: Ember, HSBC estimates

Starting with a mix of coal, nuclear, gas and hydro in 2000, gas and renewables have replaced coal over the last 10-15 years, which we expect to diminish to irrelevant levels by 2030.

Gas and renewables are pushing coal to near zero by 2030

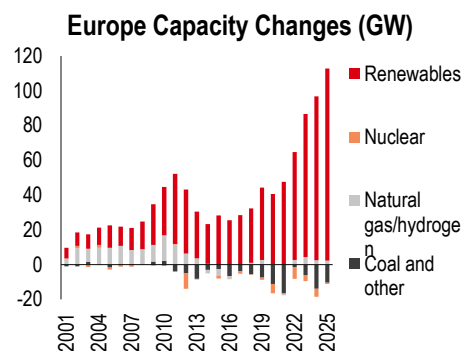


Source: Ember, HSBC estimates

We expect renewables and batteries to dominate new supply additions

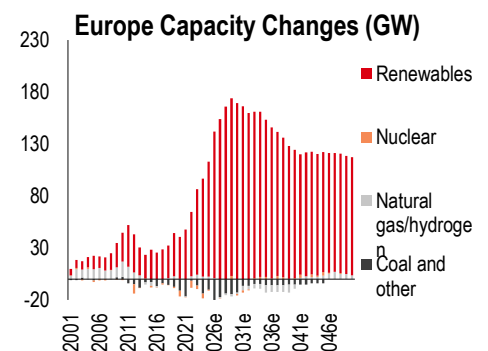
Apart from very selective amounts of net new gas and nuclear plants (10-15GW, respectively) all new capacity is renewables and batteries, which rise to additions of 170GW pa by 2030e from 110GW today and less than 50 GW in 2010.

Since 2014, there have essentially been no additions other than renewables...



Source: Ember, HSBC estimates

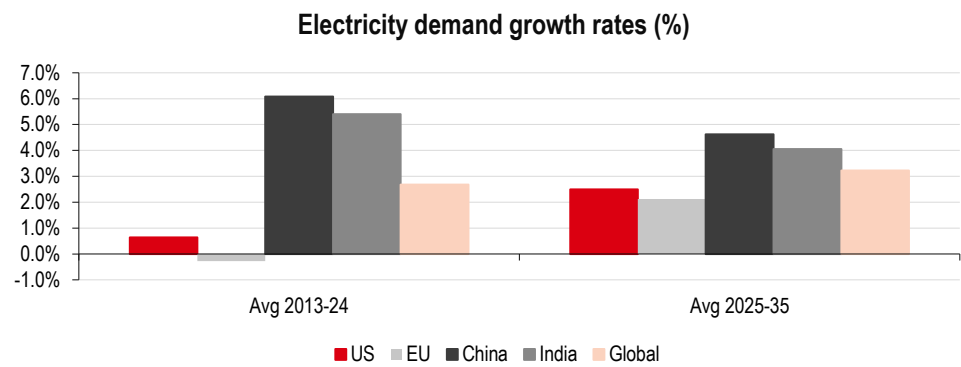
..., which will continue to be the case apart from very selective new gas/nuclear



Source: Ember, HSBC estimates

Over the last ten – or 25 – years, electricity demand has been slightly down in Europe and broadly flat in the US, relative to 1-3% annual growth expectations in Europe and 2-3% in the US in the next five years. This compares to c2-3% historical and expected global electricity demand growth as well as 5-6% annual growth rates in China and India in the last ten years and expectations of c4-5% annual growth in the next five years.

We expect growth in Europe but behind global, US and Asian growth



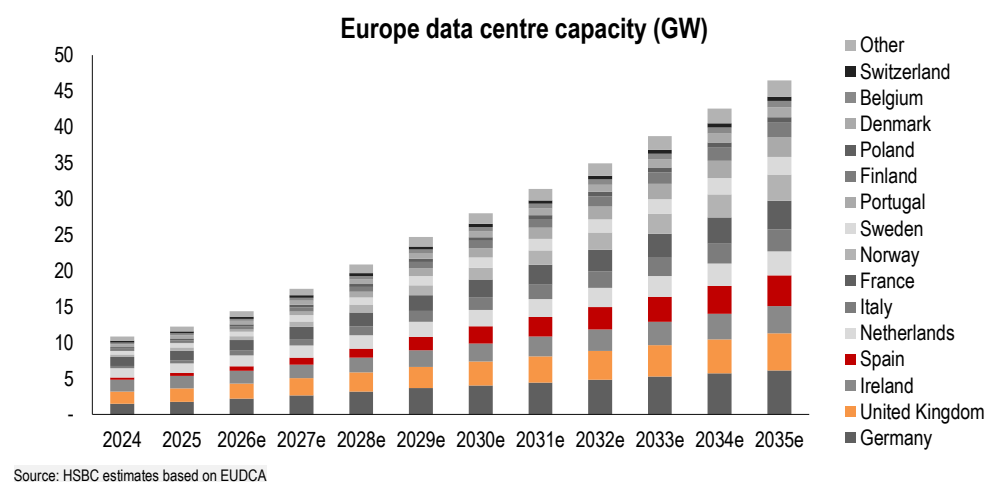
Source: Ember, HSBC estimates

Europe AI load growth deep dive

- ◆ We expect continued strong growth in tier 1 markets of Germany, the UK, the Netherlands, France and Ireland....
- ◆ ... with expansion to and strong growth in new tier 2 markets of the Nordics, the South (Iberia & Italy) or the East such as Poland
- ◆ Expectations continue to increase with 2026 forecasts for data centre capacity by 2030, 20% higher yoy

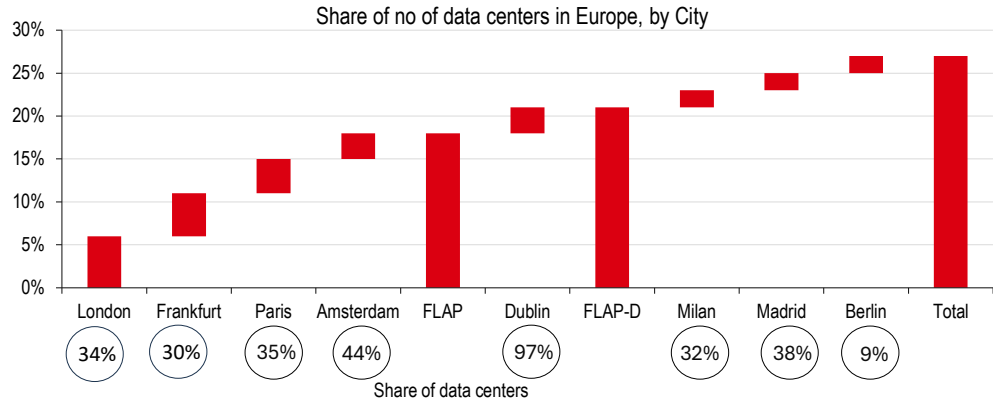
We expect data centre capacity in Europe to triple by 2031 compared to 2024 as the established tier 1 ('FLAP-D') markets – Germany/Frankfurt, the UK/London, the Netherlands/Amsterdam, France/Paris and Ireland/Dublin – continue to grow, and new tier 2 markets – such as the Nordics (Denmark, Sweden, Norway, Finland) and the South (Spain, Portugal, Italy) – emerge with strong growth.

We expect broad-based data centre growth across Europe



When initially data centre growth in Europe was centred in FLAP-D markets (Frankfurt, London, Amsterdam, Paris and Dublin) today, these markets are forecasted with continued strong growth, while other markets are emerging. This includes a broadening out of growth in Germany to Berlin and regions other than Frankfurt or the Nordics, Milan, Madrid and other regions in Europe.

Initially data centre growth in Europe was centred in FLAP-D markets (Frankfurt, London, Amsterdam, Paris and Dublin)

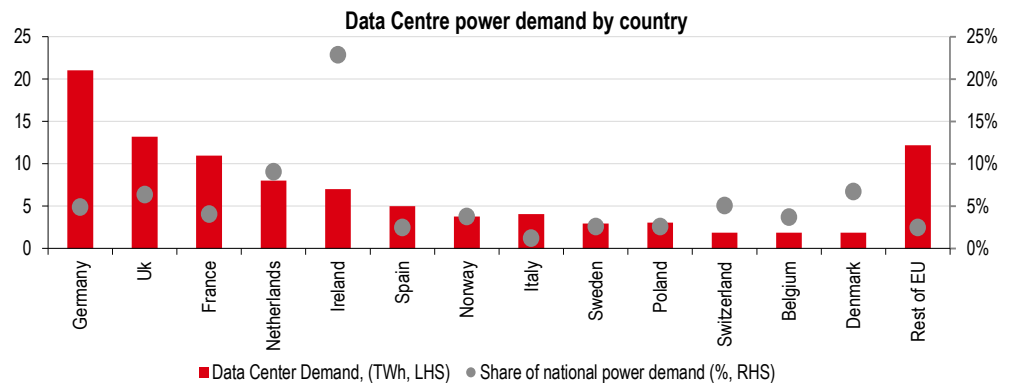


Source: CIS analysis based on DataCenterMap and Statistics

In 2025, the European Data Centre Association (EUDCA) and other forecasters debated a potential shift to the Nordics based on large project announcements as well as some signs of grid saturation around central Frankfurt, Amsterdam and Dublin. Ireland and the Netherlands stood out with the highest share of data centre power demand by country. *EUDCA's STATE OF EUROPEAN DATA CENTRES 2025* report showed that survey respondents saw access to power as the biggest challenge, yet respondents also expected strong growth in FLAPD, South, East and Nordics.

Core markets will continue to be strong, but growth is broadening out

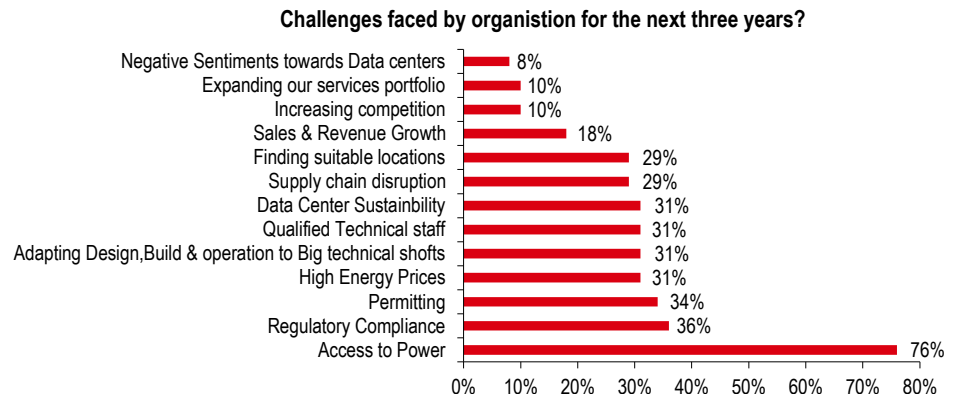
Ireland and the Netherlands stand out with the highest share of data centre power demand by country



Source: ICIS

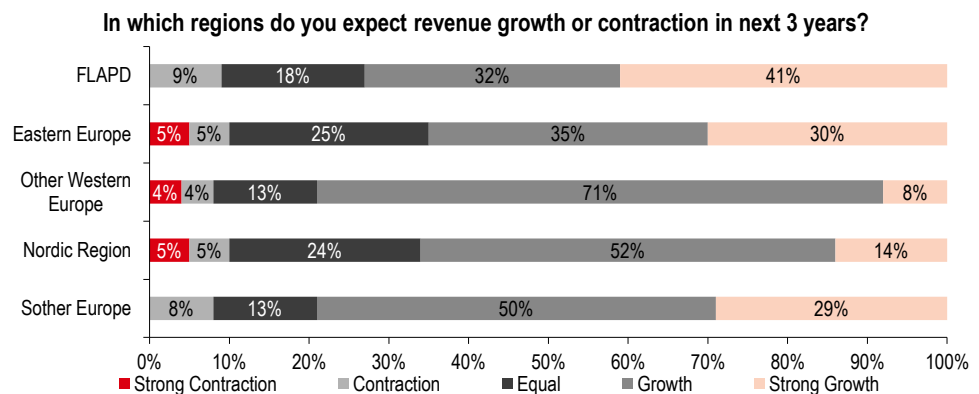
"The single most important inhibitor to colocation data centre growth is power constraint. This has a particularly big impact on the FLAPD markets. [...] In Amsterdam and Dublin, the constraints are ubiquitous: Power grid companies are unable to deliver the desired quantities, waiting lists are creating delays, and investors are directing their plans to other locations. Paris, Frankfurt, and London are also dealing with serious congestion challenges. However, these are being tackled in part through the development of data centre zones slightly outside the metropolitan areas." - EUDCA STATE OF EUROPEAN DATA CENTRES 2025

A survey in 2024 quoted access to power as the biggest challenge...



Source: EUDCA

...while respondents still expected growth or strong growth across the board



Source: EUDCA

We are still in a phase of rising applications and estimates...

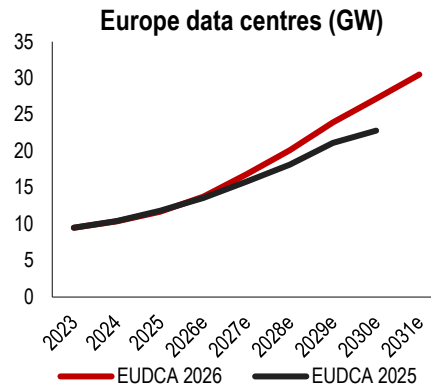
We have lifted our European electricity demand growth forecasts from 1-2% to 1-3% due to strong data centre demand

In 2026, the debate seems to have moved on to expecting strong growth across tier 1 and tier 2 markets. In its 2026 report, EUDCA forecasts strong growth across regions.

In 2026 versus 2025, EUDCA 2030 expectations increased by c20% as the annual rate of expected additions for 2027-2030e rose to 3-3.5GW (versus previously 1.5-2GW pa). We increased our estimates on the back of this analysis. We previously expected European electricity demand to grow by 1-2% and now forecast annual growth of 1-3%.

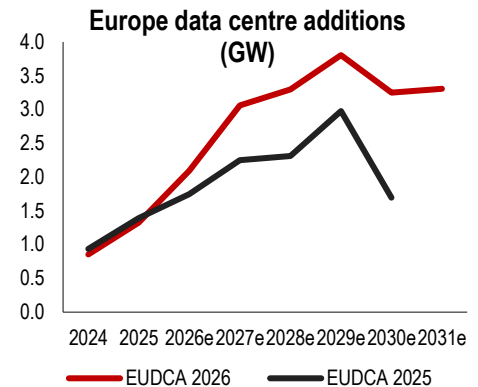
We also note that the rate of additions flattening out was a typical forecasting error. For example, the same happened when mainstream forecasters underestimated solar growth during the steep part of the S-Curve. Unless there are real long-term constraints – not execution constraint that can be managed over some years – it would seem more likely than not that addition would continue to rise rather than tail off.

In 2026 versus 2025, EUDCA 2030 expectations increased by c20% ...



Source: Ember, HSBC estimates

..., as the annual rate of expected additions for 2027-2030e rose to 3-3.5GW (versus previously 1-5-2GW pa).



Source: Ember, HSBC estimates

... while utilities work on sorting out queues

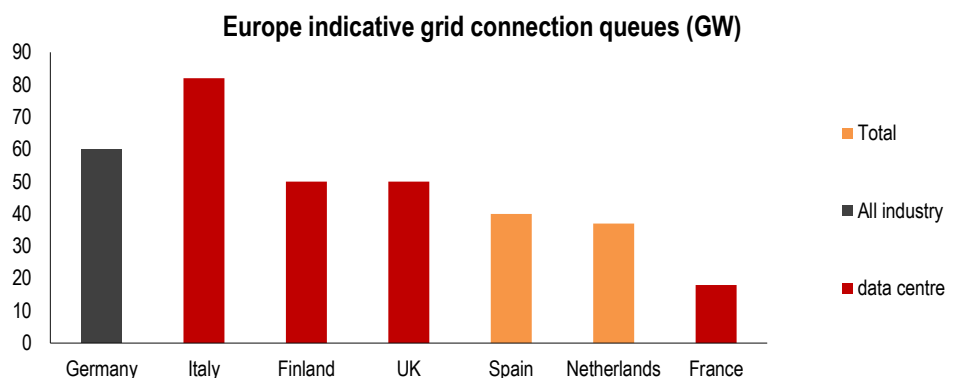
On the generation as well as demand side, grid applications typically show eyewatering numbers in every country in Europe as well as the US. However, this is due to the absence of material cost and thus overapplications and speculation. Utility CEOs and industry experts typically see these numbers as vastly inflated and estimate 10-20% of the absolute gross numbers as a more realistic order of magnitude.

Grid applications for data centres could be inflated by 5-10x

The exhibit below shows broad references of demand queues for selected European markets. Given differing methodologies, we would not suggest a comparison between countries on this basis. The scale is evidently huge with tens of gigawatts for most markets relative to annual addition expectations of c3GW on average in the next five years or forecasts of total capacity of c30GW by 2031e for Europe.

For Sweden, Germany and France, applications are broadly at 30-70% of peak demand, while for Italy, the UK, Spain it is 1-1.5x peak demand. For Finland, the Netherlands or Denmark, applications equate to 2-3x peak demand.

Interconnection queues are huge with tens of gigawatts per market



Source: Regulators and companies

The UK is proposing grid fees to secure a connection offer to help reduce speculative applications

How does the sector manage queues? In broad strokes, system operators will put up gates, such as quality requirements (e.g., demonstration of land, capital or contracts) or up-front cost/binding commitments and move to batch processing or awarding projects based on set criteria rather than a simple first come first served.

For example, the proposed UK data centre connection reforms are designed to tackle speculative DC projects that have complicated the connection queue process. Under the consultation, Ofgem has suggested that data centre developers will pay a refundable (if the project is energised) GBP200-700k per MW fee to secure a connection offer. This fee is forfeited if the project drops out.

In [new era of load growth in the US](#) we outline some examples and explain the role of data centre tariffs.

Beneficiaries? Renewables and power grids

- ◆ With Europe having started investments ahead of demand, power systems are generally ready for growth – DC or otherwise
- ◆ While the entire power value chain benefits in the US from demand...
- ◆ ...with strong energy independence and transition goals, beneficiaries of growth in Europe skew to renewables and power grids

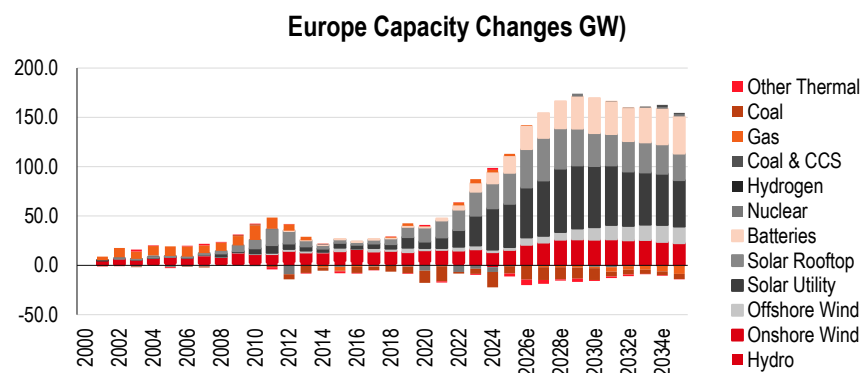
Renewables boom supported

To achieve decarbonisation/energy independence and to meet growth, Europe is already investing heavily in electricity networks and renewables. We expect wind, solar and BESS capacity additions to rise from c100GW per annum in 2024/25 to c170GW by 2030.

We forecast that the European offshore wind market will mature and that additions will rise from single digits to >15GW per annum in the mid 2030s. In 2026, we expect large offshore wind auctions in the UK and France as well as smaller auctions in Denmark, the Netherlands and Belgium. By 2027, we believe Germany will return to the market with new auctions and momentum.

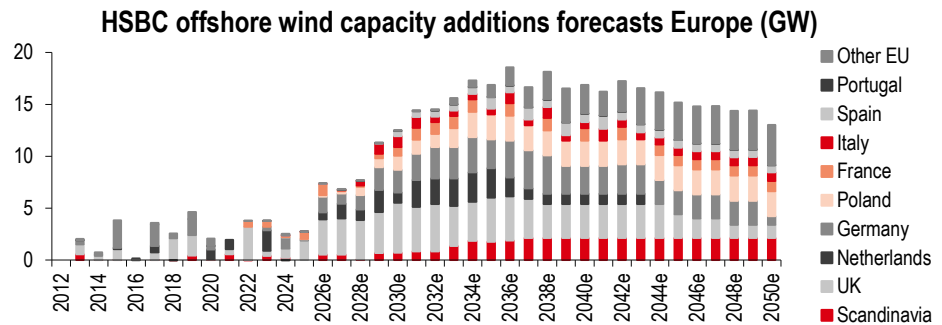
We believe that growth from data centres supports existing ambitions with incremental upside potential. Recent policy direction in Europe has been to double down on green investments as the Middle East crisis of 2026 is the second fossil fuel crisis in four years (after the 2022/23 European gas crisis) that highlights pricing, geopolitical and reliability risks of oil and gas for the EU.

...with capacity increase of up to c170GW per year



Source: Ember, HSBC estimates

We forecast European offshore wind additions to rise to >15GW per annum by 2030s



Europe at the forefront of the BESS storage boom

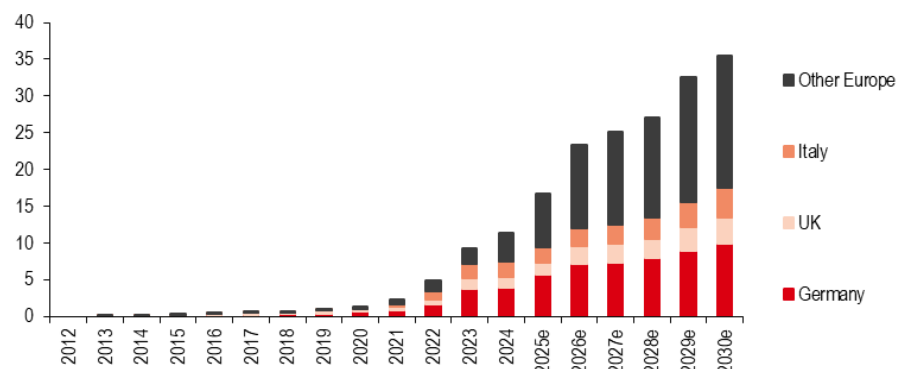
Europe makes up more than half of all global BESS installations. Since 2020, European utility scale BESS capacity has increased 11x from 4GW to 46GW in 2025. We expect this strong growth to continue: European BESS capacity to rise 3x by 2030 and reach 365GW by 2035 – a 23% CAGR.

We expect European BESS capacity to rise 3x by 2030

The policy environment is positive due to continued focus on decarbonisation ambitions with a greater emphasis on greater energy security providing an additional tailwind. Infrastructure investment across renewables, grid modernisation and end-consumer demand are all supportive of growth in battery capacity. While the broader trends remain a constant, the maturity and market potential for BESS vary significantly from country to country. The UK is one of the most advanced markets for BESS globally, Germany has a large residential capacity base, Spain and France have had limited requirement for BESS thus far, while Italy has taken the first step towards capacity auction-based growth. In addition, we note there are selective growth opportunities in other markets.

The UK is one of the most advanced markets for BESS global, Germany has a large residential capacity base, Spain and France have had limited requirement for BESS thus far, while Italy has taken the first step towards capacity auction-based growth

HSBC storage additions (GW) Europe



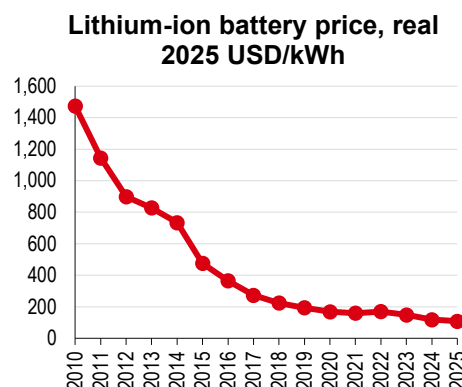
Global BESS dynamics

In the past five years, BESS have emerged from a rounding error to become a major area of capacity growth and investment for utilities and developers. Over 2015-19, global installations of battery storage totalled c10GW, while in 2020-24 they increased to 146GW, and in the next five years we estimate cumulative installations of slightly under 600GW. On a CAGR basis over the past 10 years, BESS installations have grown at 2x the rate of solar PV installations – the fastest growing generation technology. There are three key reasons for the rapid growth and why we expect this to continue:

1. **Cost competitive:** Lithium-ion battery pack unit costs have decreased 77% in the past decade and 27% from 2023 to 2025 due to a substantial build-out of manufacturing capacity in China and improved energy density. Standalone storage costs have fallen further (-45% YoY in 2025), due to oversupply and technology improvements. In many key markets, solar plus storage is competitive with fossil fuels and nuclear technologies. We expect this trend to continue, with battery pack prices falling 42% by 2030.
2. **Rise of renewables:** Energy storage/flexibility is highly complementary to wind and solar technologies. Put simply, energy storage enables a better matching of electricity supply and demand. Higher renewable penetration creates arbitrage opportunities for standalone systems and improves capture rates for developers through co-location. BESS can also provide vital grid firming, load shifting, and system service capabilities.
3. **Policy momentum improving:** System operators have been improving software and broadening remuneration options for BESS, having historically favoured fossil fuel alternatives. Capacity markets are expanding in select markets and can provide an underpin. In Texas, recent legislation to reduce the impact of large-load consumers, such as data centres on the grid, are supportive for BESS solutions. Broader grid connection and permitting reforms are directly/indirectly helpful for BESS build-out.

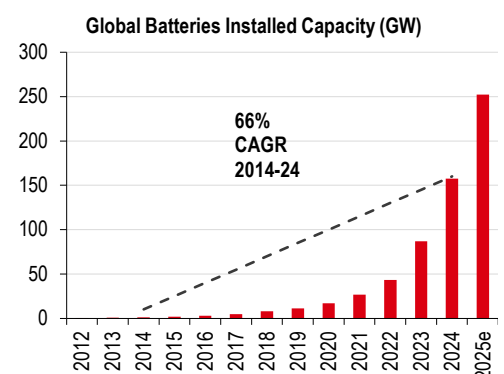
For more detail: [Global Renewables & Utilities: Energy storage the great enabler](#) (23 Feb 2026).

Over 90% decline in battery pack costs has drastically improved the economics...



Source: BNEF

...helping to drive a 77% increase in global capacity additions since 2020



Source: BNEF, EMBER, HSBC estimates

Power networks see rising investments

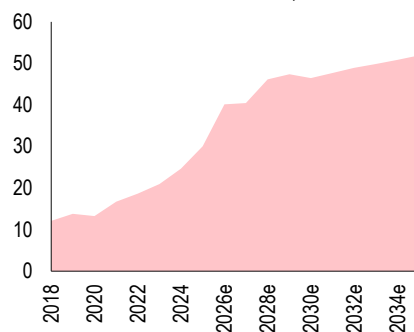
Investment in European electricity networks are rising 3-4x

Investment needs in European electricity networks are tripling to quadrupling for 2030-40 compared to 2018-20, to connect renewables, enable new demand, such as data centres and electric vehicles, digitalise networks and to increase resilience to extreme weather, in addition to ongoing maintenance investment.

Electric/regulated investment is the....

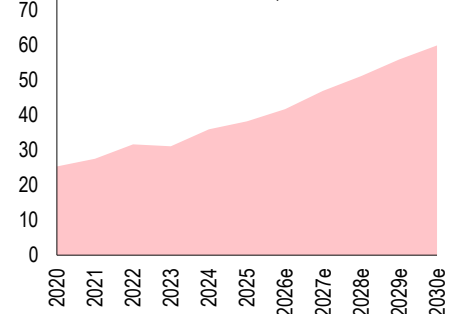
... largest structural growth driver

Electricity grid & CAPEX selected EU utilities & NextEra, EURbn



Source: Company data, HSBC estimate

Selected EU utilities & NextEra network EBITDA, EURbn

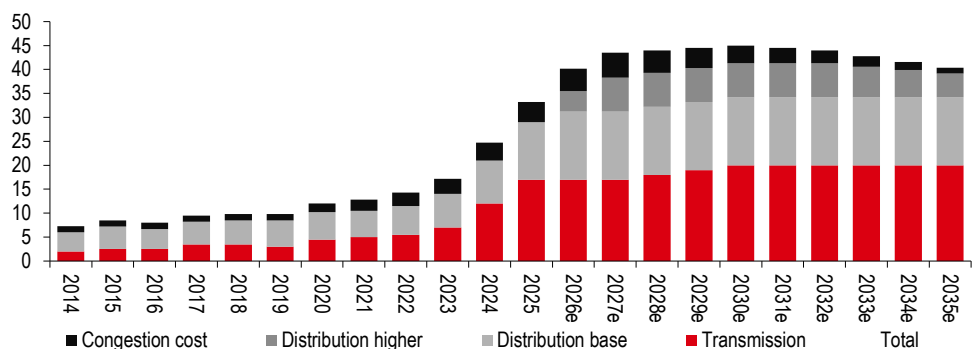


Source: Company data, HSBC estimate

In Germany, we expect annual grid investments to rise c4x for 2026e-30e versus 2020. Updated network investment plans expected in 2026 could increase these levels further and the German government in July 2026 highlighted a positive stance towards the segment with ambitions to accelerate planning & permitting and a possibly of raising investments.

In Germany, annual grid investments are rising c4x for 2026e-30e versus 2020

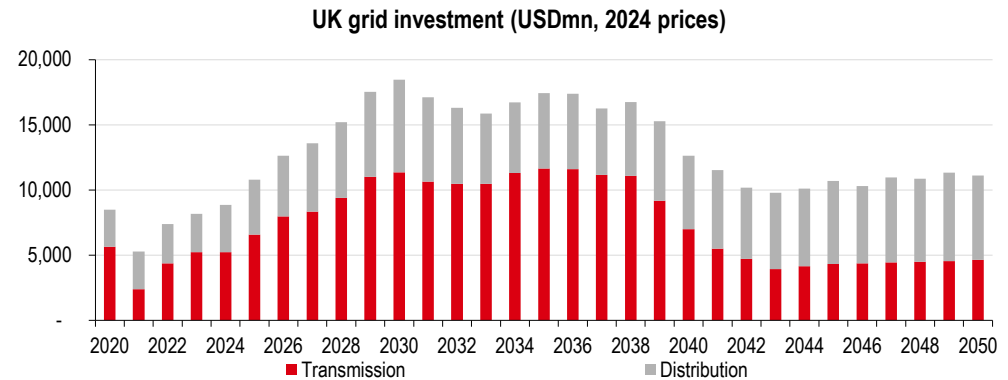
Germany grid investment and congestion cost (EURbn)



Source: Ember, HSBC estimates

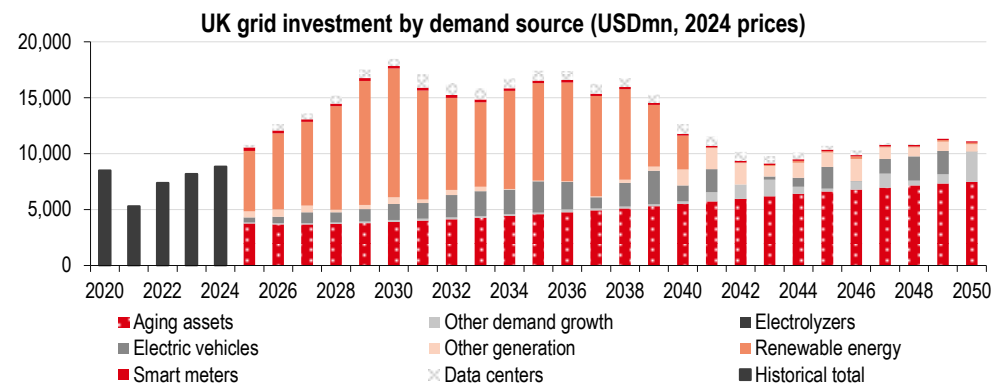
We expect UK grid investment to grow until at least until 2030, and at a minimum plateau thereafter. The step up in UK grid investment is among the highest in Europe (2022-27 CAGR: 28%) reflecting underinvestment in the 2010s, as well as an ambitions to rewire the power sector and decarbonise through predominantly offshore wind. Over the next five years, replacement of older infrastructure and connecting renewables growth will be a driver behind c80% of total grid capex. Data centres will remain a relatively small portion of capex – reaching 5-7% in the 2030s, according to BNEF.

UK grid investment by 2030 is expected to increase 3-4x vs 2020-21 levels



Source: BNEF

The bulk of UK grid investments are going towards new renewable connections and replacing/upgrading existing assets.



Source: BNEF

Nuclear, gas and coal largely unaffected by AI power demand

We don't believe that data centres will materially change the coal, gas or nuclear outlook for Europe. Europe has invested in renewables and electricity grids ahead of growth for over a decade while electricity demand has fallen in the last five years and is slower to step up compared to the US in coming years on our numbers.

We don't expect a data centre boom to materially change the outlook for nuclear, gas and coal

We believe that current power investment plans (largely green) in Europe can accommodate expected data centre growth without major adjustments. In a nutshell, we see demand and supply well balanced in Europe and thus no overhang of demand that creates material pressure for additional generation. If a case of material demand overhang were to emerge in Europe, given strong low carbon commitments in power, we would assume upside for renewables and electricity grids first before gas, nuclear and coal.

Europe's power system demand-supply balance is materially different from our view of the US power system, where fewer investments in growth/upgrades in the past decade face immediate higher demand in a backdrop of lower net-zero commitments (compared to Europe). As network and nuclear scale ups take time and can be expensive in the case of nuclear, gas generation is

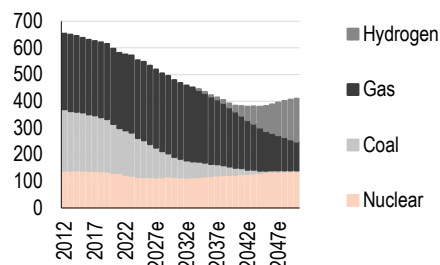
an affordable and convenient outlet for excess demand in the US (a country with its own gas resources), but not in Europe.

In Europe, we model coal closures broadly by 2040, replacement of nuclear and moderate growth by 2050, with part of the gas fleet eventually upgraded to hydrogen long-term. Coal output disappears by 2040e, the role of gas in the European system moves to back-up and may be upgraded to hydrogen long term. Load from nuclear rises moderately back to 2012 levels long term.

In Europe, we model coal closures broadly by 2040, replacement of nuclear and moderate growth by 2050, with part of the gas fleet eventually upgraded to hydrogen long term.

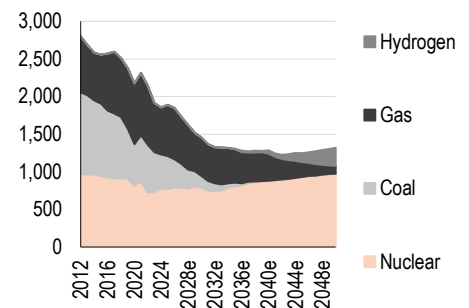
Coal output disappears by 2040e, the role of gas in the system moves to back-up and may be upgraded to hydrogen long term. Load from nuclear rises moderately back to 2012 levels long term.

Europe Capacity selected technologies (GW)



Source: Ember, HSBC estimates

Europe Load (TWh)



Source: Ember, HSBC estimates

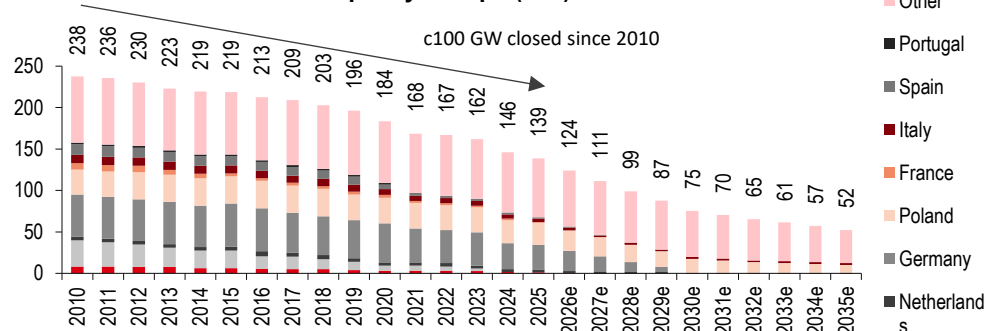
Coal phase out plans going ahead

We observe that major economies of Europe continue to execute against their coal closure plans. The UK, as a prominent example, has already closed c32GW of coal over the last ten years and countries like Spain or Italy are on the same path.

Today, large parts of European coal sit in Germany, Poland and Eastern Europe and we expect a large wave (over 60GW) of further closures in the next five years. We believe that German coal closure plans by 2030 (pulled forward from 2038) are on track and we expect the remaining 30GW of the German coal fleet to close by 2030/31e. We also expect further closures in Poland and Eastern Europe. Thus, we forecast European coal capacity to be down to less than 80GW by 2030e, falling further to c50GW by 2035e from c140GW today.

C100GW of coal has closed since 2010 in Europe with a further c75GW in the next five years

Coal capacity Europe (GW)



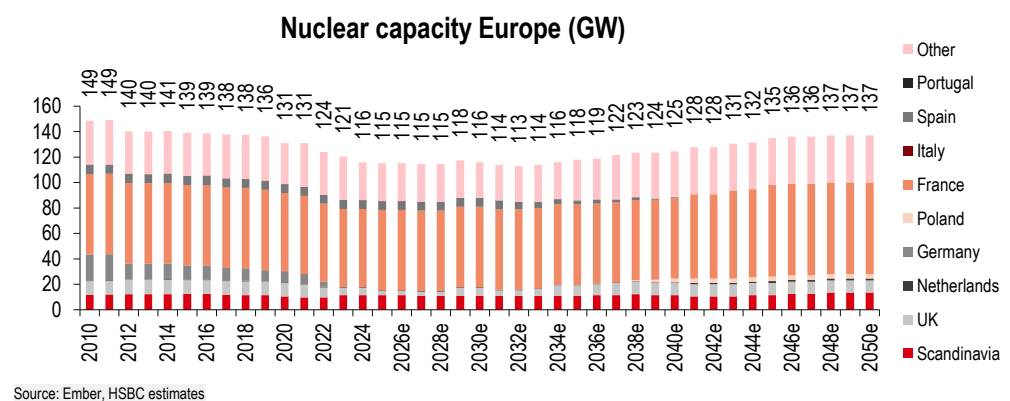
Source: Ember, HSBC estimates

One positive for owners of Coal assets from AI and data centres is the possibility of using former coal sites for data centres and utilities such as RWE have already announced deals. Deals can take various forms: the site only, site plus upgrade work or a full service PPA that also includes providing generation.

Maintaining a nuclear industrial base

We believe that sentiment toward nuclear in Europe has become incrementally more positive in the last five years driven by energy independence goals and political intervention to maintain a nuclear industrial base; AI & data centres play a minor role. Instead of closures, we see incremental life extensions, such as in Belgium and Spain. Beyond 2035, we also model some new addition as mainly French, British and Eastern European plans for new large and small modular reactors might come online. By 2050, we model c137GW of nuclear capacity in Europe, broadly in line with 2012-2019 levels and compared to c115GW today. In the 2050s, the French nuclear fleet would reach 80 years of life and continued construction activity would be needed to replace the existing fleet.

Nuclear growth beyond 2035



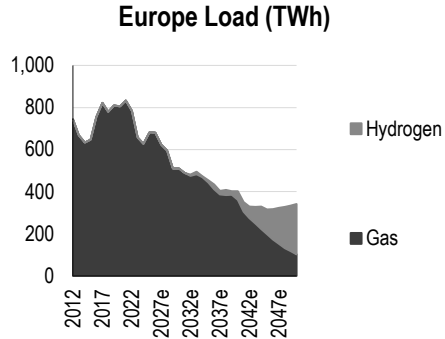
Gas generation will remain for back-up, but not for new growth

Gas/hydrogen for back-up not for growth

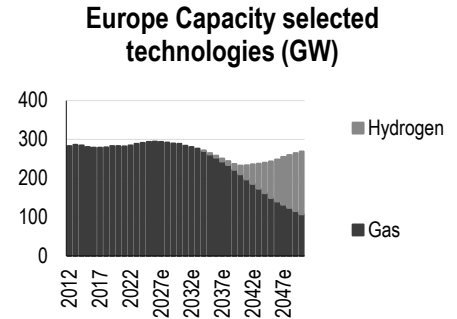
As the share of renewables in European power systems rises, gas/hydrogen is the back-up technology that complements renewables. In the long-run, part of its role, such as peaking capacity, should be addressed by batteries or flexible demand while the need to provide back-up over days such as during dark lulls will continue to require large-scale CCGT assets. Eventually, depending on European subsidies and ambitions to remove gas from the power system, gas might be replaced by hydrogen in the long term.

Over the last three years, we believe the European view on gas/hydrogen has changed as follows: Ambitions to replace gas with hydrogen have been pushed back and new gas additions have become more selective, although recent UK capacity auctions have awarded gas upgrades/extensions. For example, German capacity auctions, which are three to four years behind schedule, were initially expected to support up to 30GW of new gas. The final design of the auctions looks for less overall capacity with only 9GW specifically reserved for gas and the remainder technology neutral. Like nuclear or coal, we believe data centre demand plays a minor role in the outlook for gas capacity in Europe.

Europe continues to see gas as back-up



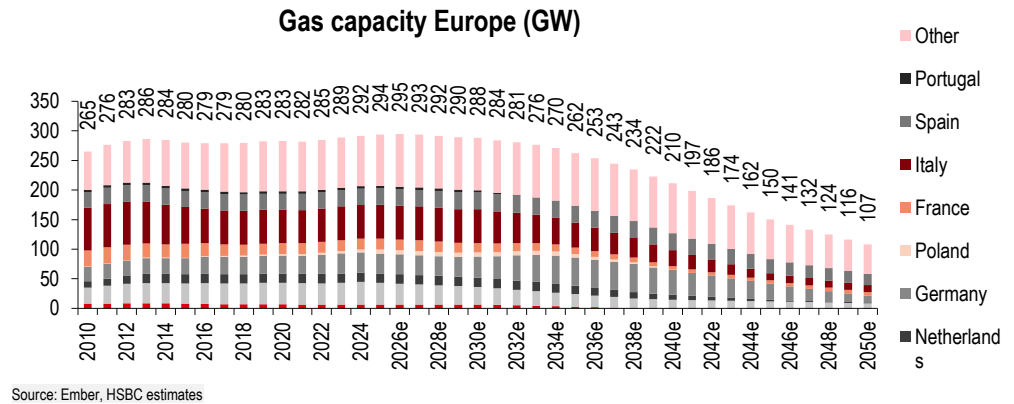
Ambitions to replace gas with hydrogen have been pushed back



We model European gas generation capacity to be broadly stable over the next few years

We model overall European gas generation capacity to be broadly stable in next few years, no major development for growth. We model the overall gas/hydrogen capacity as broadly stable long-term and the outlook for gas specifically is mainly dependent on political ambitions to replace gas eventually with hydrogen or not.

Gas: the outlook for gas specifically mainly depends on political ambitions to replace gas eventually with hydrogen or not



Affordability versus strategic industry

- ◆ Europe is welcoming data centres, with governments viewing it as a strategic industry
- ◆ Energy affordability in the EU means energy independence via homegrown renewables/grid investments
- ◆ Green investment boom for over a decade has created headroom for data centre growth at affordable prices

European governments see AI and data centres as a strategic opportunity or even necessity and overall welcome the rise of this new industry to offer jobs and investments. Governments are concerned with energy independence as much as energy affordability; yet in Europe these goals go hand in hand while AI & data centres do not feature prominently in the debate; the focus is on fossil fuels versus clean tech.

As Europe has planned for top-down electricity demand growth and invested accordingly over the past decade, the electricity supply demand balance is more comfortable in Europe compared to the US. In our view this leads to lower execution risk: Generation and network expansion activities have already scaled up to meet growth from all sources, including data centres and AI.

Ambitions to host strategic industry

Europe and individual European countries have data centre targets of doubling and tripling capacity by 2030. On our forecasts, Europe is on track to broadly achieve these ambitions. Some details below.

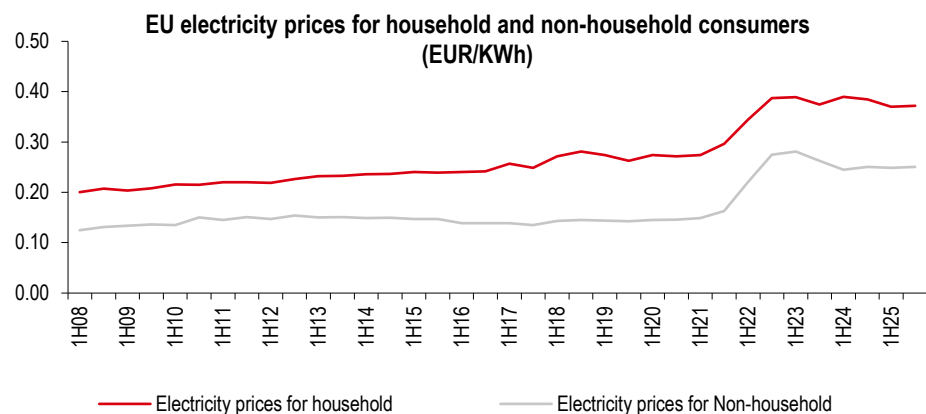
- ◆ Germany launched its National Data Centre Strategy in March 2026. It aims to double DC capacity by 2030, while increasing AI and high-performance compute capacity 4x over the same period. The strategy follows on from the 2025 legislation that formally moved DCs to critical infrastructure status. Policy proposals include faster grid connections, accelerated planning and permitting procedures, and supporting AI industries and technology.
- ◆ France's National Strategy for AI is embedded in the broader France 2030 investment plan, which includes EUR109bn of investment directed towards AI infrastructure. Private investment is a key component. Softbank has pledged EUR75bn to develop 5GW of AI DC capacity, with an initial investment of EUR45bn by 2031.
- ◆ AI is a core pillar of the UK's industrial strategy to support frontier industries. The primary strategic initiative is being delivered through AI Growth Zones (AIGZ) – five designed areas that will have targeted policy (streamlined permitting, discounted electricity etc) to

encourage large-scale data centre developments. The previous government committed GBP2.5bn in direct investment to AIGZs. This said, development progress has thus far been slow, in part due to high eligibility criteria such as projects having access to an existing connection of 500MW, as well as political uncertainty.

Affordability: Independence at no extra cost

Europeans had their energy affordability shock in 2022 driven by oil and gas prices not by renewables or data centres. In 2026, the Middle East crisis is another reminder that for fossil fuel importing nations like Europe, fossil fuels are a much larger driver of energy affordability than renewables or electricity network investments or higher electricity demand from data centres. In 2022, European power prices broadly doubled driven by higher gas prices after Russia's invasion of Ukraine and the reduction of Russian pipeline gas flows to Europe. Since 2022, power prices have been stable to trending downward.

EU Retail and industry electricity prices



Source: EU statistics

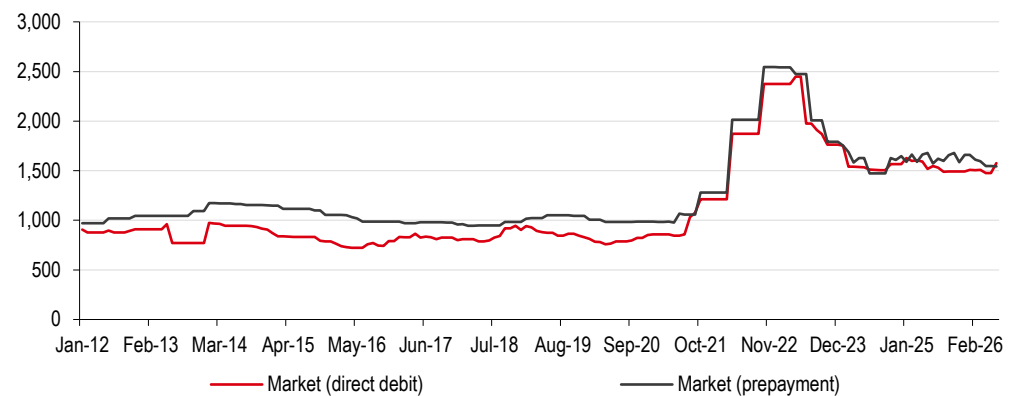
Example: Affordability a key focus in the UK

The cost of living continues to be a main debate for UK politics, and energy affordability is a key focus. Compared to pre-Ukraine war levels, the household dual fuel (electricity and gas) energy bills are up c50% and the ongoing conflict in the Middle East has added further upwards pressure to wholesale gas and to a lesser extent, also electricity prices in the short term. Policy initiatives are focused on reducing the cost of electricity, rather than the gas portion of the standard bill, as part of the strategic aim to electrify the economy and minimise reliance on more volatile fossil fuels. Furthermore, stimulating and expanding electricity demand plays a significant role in bringing down the fixed costs of network and generation investments, and therefore the unit cost of electricity for consumers. The main levers used thus far have been to move policy costs, such as RO (renewable obligations) subsidies to general taxation, as well as removing VAT from electricity bills and the Carbon Price Support by April 2028.

In the medium term, there are several competing forces for dual fuel bill costs. The price of natural gas will continue to be a key variable for both the gas and electricity components. Assuming an end to the hostilities in the Middle East, we would expect a supply glut to emerge later this decade, putting downwards pressure on wholesale prices and therefore bills. We believe the sharp uptake in BESS will be deflationary for electricity bills, reducing reliance on gas peakers and lowering capacity payments. On the other hand, network and policy costs will step up over the next five years, and while we believe investments in renewables and

network infrastructure will be supportive for bills (lower wholesale and constraint costs), the timing of these benefits remains uncertain. Another variable is the size of the upcoming renewables auctions. If AR8, AR9 and AR10 are a similar magnitude to AR7, then additional grid investment might have to be brought forward.

UK dual fuel bill (GBP/year): average fuel cost up 50% vs pre-Ukraine crisis levels



Source: Ofgem

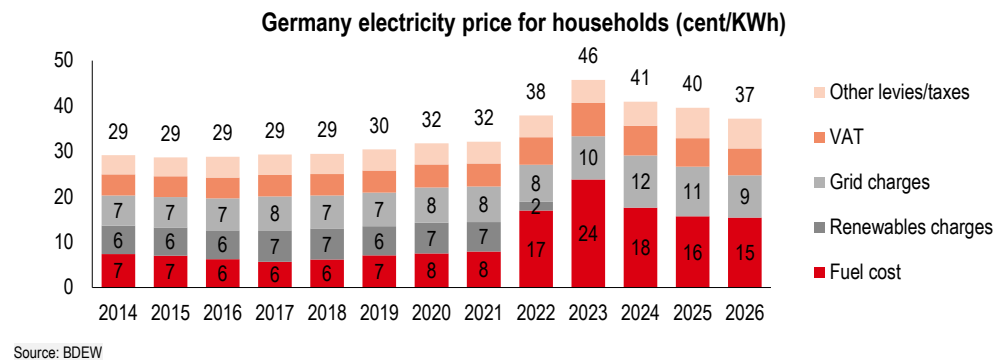
Example: Affordability in Germany

We outline German bills as a second example. Gas and to a lesser extent electricity prices (buffered by renewables generation) in Germany spiked in 2022/23s. With higher wholesale power prices, German state renewables subsidies, which are essentially a top up to power prices, became obsolete during the crisis and the government moved the much-reduced burden to general taxation.

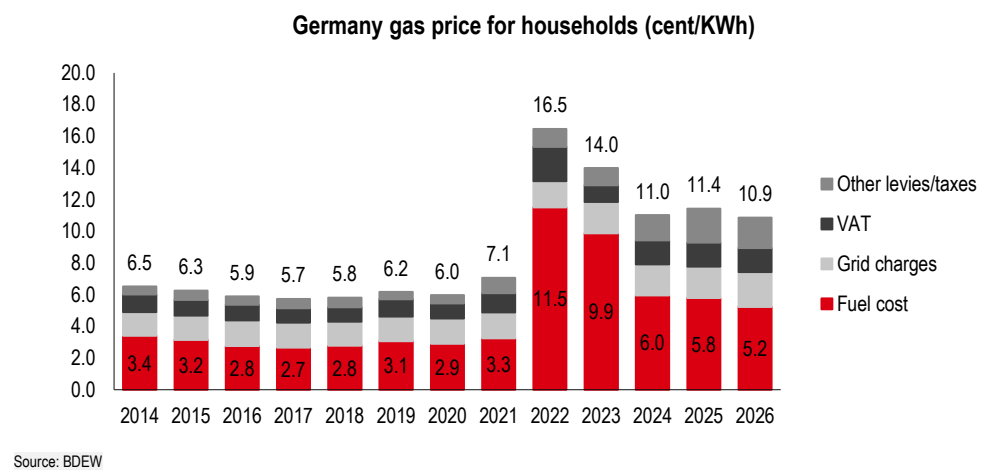
A gas shock that sent household gas prices from c6 cent/KWh in 2020 to c16 cent/KWh in 2022 and c11 cent/KWh in 2026, only increased retail electricity prices from c32 cent/MWh in 2020 to c46 cent/KWh in 2022 and c37 cent/KWh in 2026.

2026 vs 2020 in Germany: Industrial electricity prices are broadly unchanged, while household electricity prices are 16% higher and household gas prices are 83% higher.

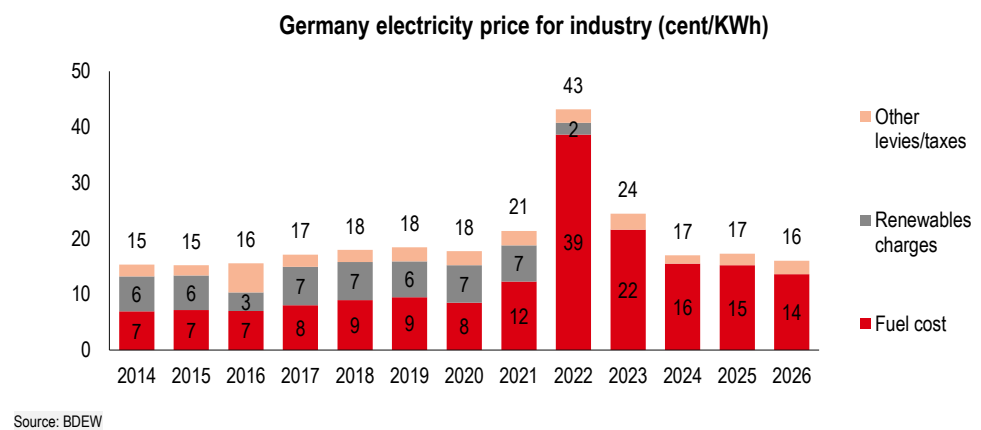
German residential electricity prices peaked in 2023



... as did household gas prices



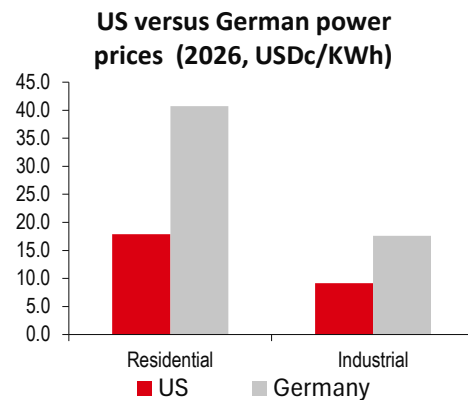
Industrial electricity prices are back down to 2014-2018 levels.



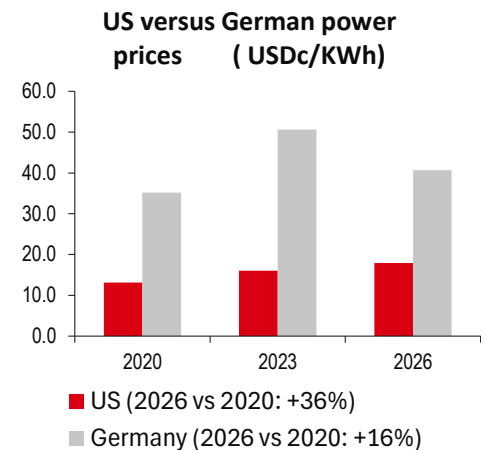
How does this compare to the US? Unsurprisingly, with much higher gas costs and with gas prices influencing power prices, German power prices are structurally more than double US prices. Yet surprisingly, price increases – despite a gas crisis in Europe – have been higher in the US over the last five years. US power prices are up 36% in 2026 versus 2020, while German retail power prices are up 16% in 2026 versus 2020.

Europe continues to see gas as back-up

Ambitions to replace gas with hydrogen have been pushed back



Source: EIA, BDEW



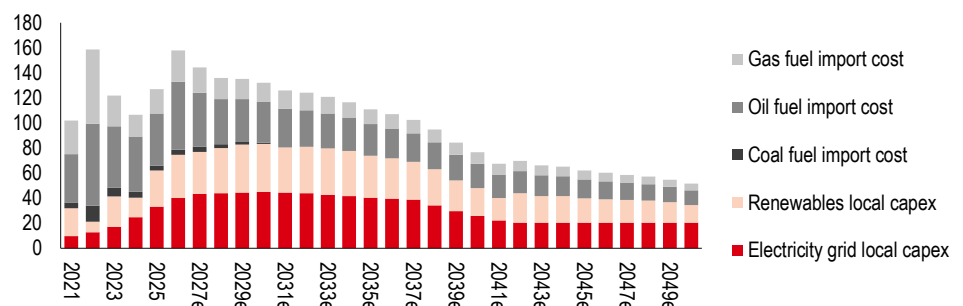
Source: EIA, BDEW

European gas costs are 3-5x the US, incentivising the move towards energy independence

As a fossil fuel importer, Europe's energy strategy has been markedly different from the US over the last decade. At 3-5x the US gas cost and with rapidly falling clean technology costs, investing in electricity (renewables and networks) is simply at par with the cost of importing fossil fuels for Europe. In addition, there are the added benefits of rising energy independence. Below we outline an illustrative example for Germany: Germany pays a huge bill for fossil fuel imports which spiked in 2022/23, remained elevated post crisis and faces upward pressure again in 2026. Investments in electricity grids and renewables have already risen materially and are close to their peak. Due to timing, paying ahead for electricity investments while still paying for fossil fuels, Germany's energy cost could already be at a peak: Going forward, the volume and with it the cost of fossil fuel imports should fall, while investments in electricity do not rise that much more and eventually should fall on lower clean technology costs and lower volumes with clean power targets achieved by 2035e.

German energy cost already at peak?

Germany: from fossil fuel imports & dependency to local investments & independence (EURbn)



Source: Oeko-Institut e.V. "Kosten der Energieimporte nach Deutschland und Europa"; BMWK, IMK, HSBC analysis and estimates

In summary, this backdrop has two consequences in our view.

- ◆ First, power prices in Europe are not expected to rise materially and structurally from the raised post 2022/23 levels. Cost increases in grids and renewables could be offset by lower wholesale power prices, lower taxes and lower congestion charges. If European governments raise affordability concerns it is typically linked to electricity prices relative to fossil fuel prices. It could be an intention to lower taxes or move charges from electricity bills to general taxation to lower the bill and thus help to incentivise faster switching to electricity. It could also be to further break the linkage between gas and electricity prices, for example by keeping renewables on PPAs.
- ◆ Second, changes to power prices and concerns regarding affordability in Europe are typically linked to fossil fuels and energy transition/independence investments. Data centres and AI do not feature prominently in the European debate on power prices and affordability.

Execution risk

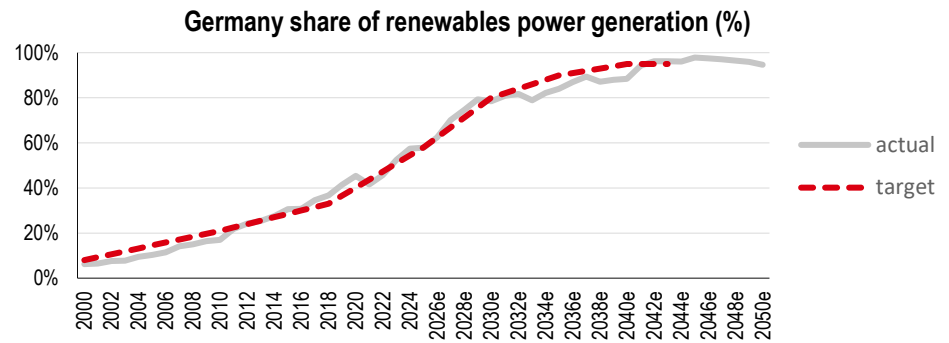
As Europe has planned for top-down electricity demand growth and invested accordingly over the past decade, while electricity demand has fallen (versus plans of growth), the electricity supply-demand balance is more comfortable in Europe compared to the US. In our view this leads to lower execution risk: Generation and network expansion activities have already scaled up to meet growth from all sources, including data centres and AI.

UK: strong momentum. Since Labour's election in July 2024, the political and regulatory backdrop for UK electric utilities has been one of the most supportive in decades. Ambitions to decarbonise the power system have accelerated, confidence in UK renewables auctions has recovered and electricity infrastructure is at the centre of the economic growth agenda. Government recommendations are for a more strategically minded regulator with statutory duties to directly support net zero, economic growth, and the UK's Clean Power 2030 targets – better aligning regulatory and political outcomes. Top-down and strategic planning provides visibility for infrastructure projects that extends well into the 2030s, helping developers to secure procurement, labour and permits well in advance.

Nordics: cheap and clean energy. The Nordic region has the cleanest energy mix in Europe, benefiting from abundant hydro resources as well as strong nuclear and wind capacity. Electricity prices are also some of the lowest in Europe putting the region in a good position to execute on AI and data centres growth strategies. Furthermore, the Nordic power system is liberalised and has a modern, well integrated grid. Large amounts of hydro capacity provide flexibility and resilience.

Germany well under way: The decarbonisation of the German power system is well advanced and Germany's share of renewables in power has tracked targets closely for the past 25 years. Renewables investments are on track and with decades worth of planning horizons for networks, network investment plans have equally been scaled for growth for years.

German share of renewables in power has tracked targets closely for the past 25 years



Source: BDEW; HSBC analysis and estimates

Blue-sky: Could the US & Europe grow like China?

- ◆ Could AIDC demand boost US and European electricity demand growth to 5% per annum? Renewables and electric grids benefit
- ◆ We assume a preference for terrestrial AIDC, especially renewables powered, and see potential for orbital AIDC with space solar
- ◆ The gas and nuclear conundrum: Terrestrial AIDC is a tailwind, while orbital AIDC might be competition

In this chapter we focus on potential opportunities in the power sector implied by SpaceX's orbital AIDC (AI data centre) ambition stated as 100GW by 2040. We split this into five sections: 1) cost comparison between orbital and terrestrial AIDC; 2) blue-sky total AIDC demand; 3) beneficiaries of maximising terrestrial AIDC; 4) beneficiaries of orbital AIDC and 5) risks and trade-offs.

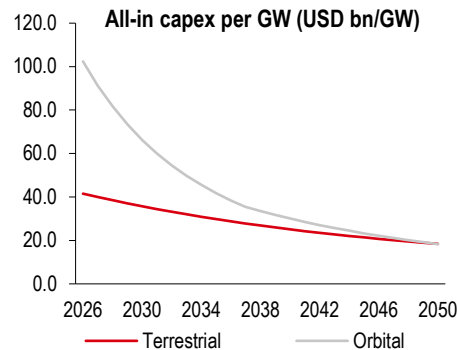
We conclude that there is potential upside to terrestrial power demand heavily skewed to the US/EU and China; high AIDC electricity demand could catapult US and European electricity demand growth to 5% per annum, levels typically associated with emerging markets. There is upside potential to the solar supply chain from orbital AIDC. While we see renewables and electricity grids as clear beneficiaries of higher terrestrial AIDC demand, the long-term outlook for nuclear and gas bears both risk and opportunities. Terrestrial AIDC could be a tailwind especially leading to further lifetime extensions of existing nuclear, while orbital AIDC might be competition for new additions.

Cost convergence could be achieved

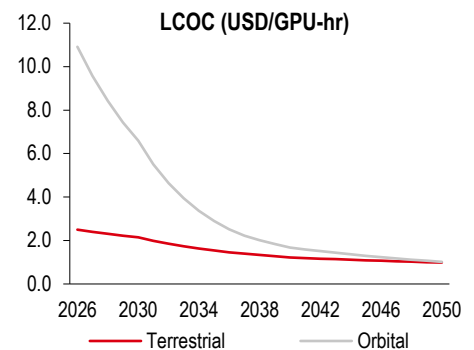
Terrestrial AIDC is cheaper compared to orbital AIDC...yet costs could converge

Today, terrestrial AIDC is expensive with investment estimates of USD40-50bn per GW. Orbital AIDC is three times more expensive on high launch cost as well as high cost for housing and radiation/redundancy.

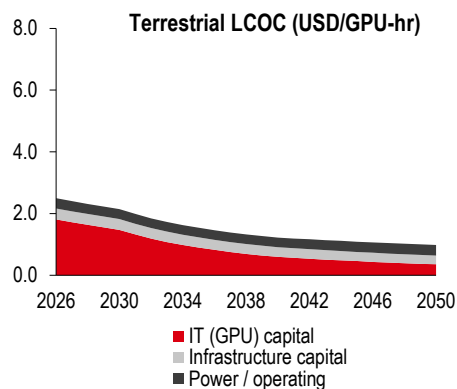
However, if orbital scales up on technology learning curves, orbital AIDC cost could fall rapidly while terrestrial costs also reduce. Long-term, the capex cost of orbital AIDC (for housing and IT) could be equal to capex cost of terrestrial AIDC (for buildings and IT). The remaining cost of electricity for terrestrial versus launching and radiation shielding for orbital also converges, leading to overall Levelized Cost of Compute (LCOC) that are similar.

As orbital and terrestrial CAPEX converges...


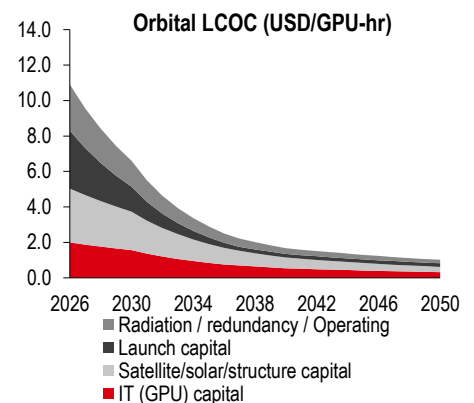
Source: SemiAnalysis, HSBC estimate

... so does LCOC. Operating costs could be similar


Source: SemiAnalysis, HSBC estimate

IT CAPEX falls for terrestrial AIDC


Source: SemiAnalysis, HSBC estimate

Orbital AIDC cost plummet across categories


Source: SemiAnalysis, HSBC estimate

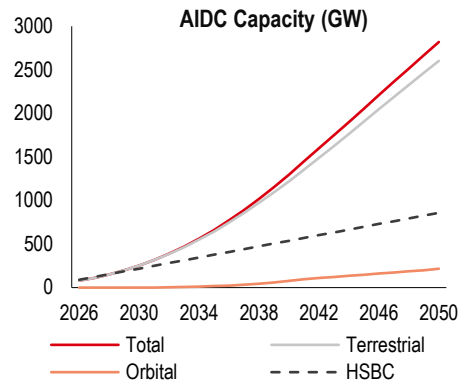
In our view, we don't necessarily need to believe that orbital becomes cheaper than terrestrial to make the business case: Perhaps it could even remain twice as expensive as terrestrial in a conservative scenario; however, costs need to come down in absolute terms and need to narrow the gap to terrestrial so as not to be prohibitive. On our numbers, this could be the case by 2035/40e.

Blue-sky: Raising AIDC demand 3x

In a blue-sky scenario, AIDC demand could rise to 2,800GW by 2050, 3x our current implied forecast

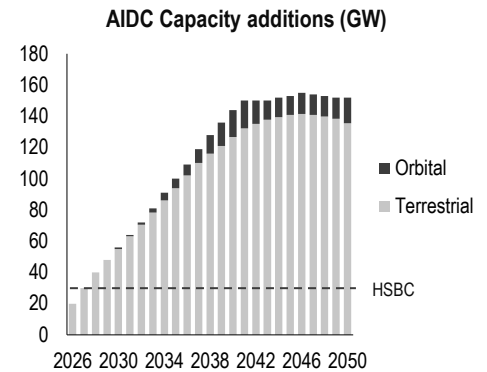
With or without orbital, at least 100GW of AIDC per annum by 2040 is a huge number and if we think of it as orbital only with terrestrial on top of it, it is even bigger. As a blue-sky scenario, we model additions ramping up to 150GW by 2045-50, with 120-140GW of terrestrial additions and 10-30GW of orbital. Cumulative capacity would rise to around 2,800GW by 2050, 3x our current implied forecast of c900GW by 2050. By 2050, annual additions would rise to 4-5x our current implied level of 30-40GW.

2,800GW AIDC capacity in 2050



Source: SemiAnalysis, HSBC estimate

AIDC capacity additions rise to 150GW by 2050



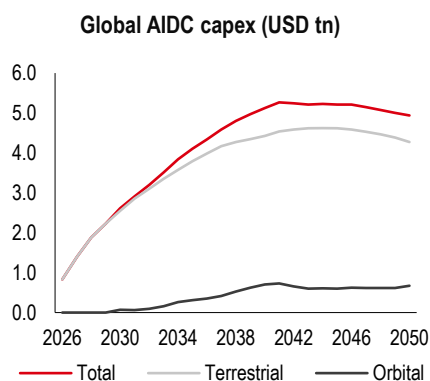
Source: SemiAnalysis, HSBC estimate

We anchor these assumptions broadly in historic IT investment trajectories and as a share of GDP. Additions combined with our cost assumptions imply new and replacement investments could rise from under USD1trn to over USD5trn by 2045-50e. Global technology capex has risen to cUSD600bn in 2026e from an average of USD50-100bn per annum in 1996-2010 (for more details, see [Show Me The Money – HSBC Global Capex Monitor: The tech supremacy](#), 1 October 2025).

Our blue-sky scenario implies AIDC capex peaks at c2% of world GDP in the 2030s

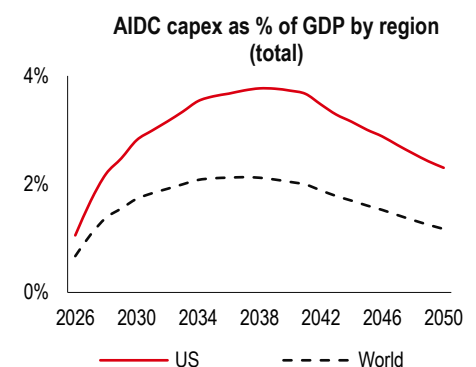
Historically, global technology investments have risen from a 0.2% share of GDP in 2005 to a 0.5% share in 2026e. Our blue-sky AIDC forecast would imply AIDC capex peaking around 2% of world GDP in 2030s before falling back to a c1% share by 2050e. With a skew of AIDC capacity to the US, it would suggest investments peaking at 4% of US GDP before falling to a 2-3% share by 2050e.

USD5trn of annual AIDC investments by 2040e...



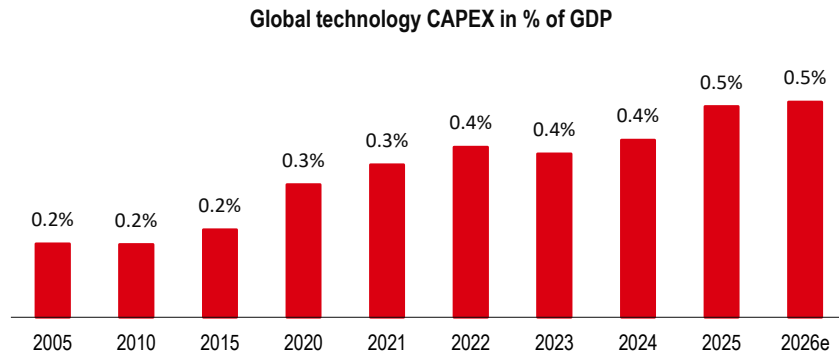
Source: SemiAnalysis, HSBC estimate

... would imply investments peaking at 2% and 4% of world and US GDP, respectively.



Source: SemiAnalysis, HSBC estimate

Historically, global technology investments have risen from a 0.2% share of GDP in 2005 to a 0.5% share in 2026e.



Source: IMF, LSEG Datastream, LSEG IBES estimates, HSBC calculations based on consensus estimates

Arguments in favour of high growth

As a disruptive technology, would AI not deploy according to an S-curve pattern of growth and adoption? We are currently still in the early stages and after the tipping point, volumes could surge. One could also see AI as a self-fulfilling prophecy, in other words a race that will be seen through to the end. It is highly unlikely that current AI leaders can afford to fall behind, let alone stop and respective governments would likely agree. This should hold both for supremacy in orbit as well as AI itself.

Critical considerations

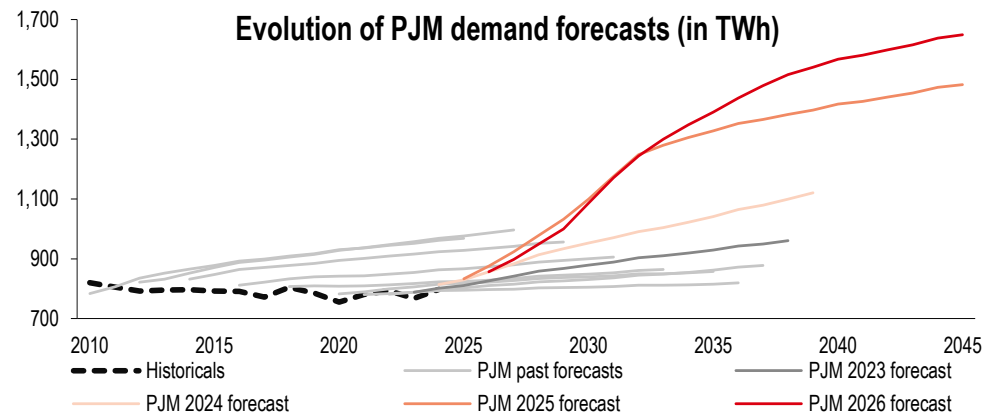
Additions in the hundreds of gigawatts seem wildly ahead of expected additions of 30-50GW pa in the next five years. It also seems to be ahead of near to medium-term capex plans of AI labs themselves.

Power perspective clearly signals growth, but how much?

We note rising estimates for AIDC power demand. For example, PJM Interconnection – a regional transmission organisation in the US hosting the largest data centre hubs – increased electricity demand growth expectations every year over the last three years, accounting for the growth in data centre connection applications.

It is too early to tell how forecasts will develop, they might well continue to rise. We note that PJM's 2025/26 curves follow an S-shape, implying current expectations of falling rather than linear or accelerating additions by 2040/45. Nevertheless, this could be attributed to poorer forecasting quality further out rather than genuine levelling off.

Power demand estimates are rising



Source: PJM Interconnection

US utilities continue to nudge demand expectations higher

Dominion Energy Virginia's forecasts show some convergence in more recent expectations, yet also potential for further increases. Dominion Energy is a utility operating within the PJM Interconnection regional grid that is close to a massive cluster of data centres in Northern Virginia that is powered primarily by Dominion Energy] and has been working with hyperscalers for over a decade.

- ◆ Growth forecasts jumped in 2022/23, from no growth expectations in 2021 to c4% annual growth outlook in 2023.
- ◆ Since then, in the 2024, 2025 and 2026 outlooks, expectations seem to converge around c5% annual growth (with forecasts inching higher more incrementally).
- ◆ Dominion's shape of growth expectations to 2045 is much more linear than PJM's forecast suggesting stable or slightly accelerating annual additions.
- ◆ However, Dominion Energy Virginia also notes that existing ESAs (Electric Service Agreements) and CLOAs (Construction Letters of Authorization) – firm capacity – exceed forecasted demand through 2045. This could imply that forecasts will rise further.
- ◆ In addition to 22GW of ESAs and CLOAs in 1Q2026, Dominion Energy Virginia has 30GW (which has risen by 3-5% in the last two quarters) of SELOAs (Substation Engineering Letter of Authorizations). As some of these convert to CLOAs and then ESAs, growth outlooks could be raised even more and eventually be in line with the blue-sky scenario of rising additions to 2045.

Maximising terrestrial potential

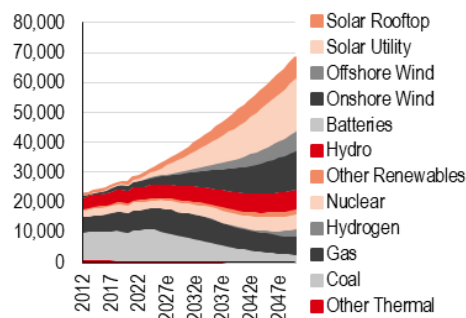
We assume that data centres would prefer terrestrial capacity due to lower latency, easier maintenance and better opportunity for scale even if the costs of orbital and terrestrial capacity converge. Therefore, our first layers to absorb higher than expected AIDC demand would be all forms of terrestrial power generation apart from coal plus the respective electricity network investments. Only if demand exceeds what can be built on Earth would orbital take off. In later sections we will challenge these assumptions and discuss trade-offs between orbital, gas and nuclear.

We think terrestrial power could support 120-140GW of AIDC capacity additions by 2045/50e. Hence for orbital to take off, we would need to believe in additions rising above that level or other strategic reasons. Space X might view terrestrial potential as more constrained than we do – let's say at 50-100GW per annum – and thus might see 50-100GW of orbital AIDC potential. This would still be in line with our overall market sizing in the blue-sky scenario of 150GW of additions by 2050e.

We currently expect global electricity demand to grow from c33PWh in 2026 to c69PWh in 2050e, broadly doubling. Terrestrial AIDC rising to c2.8TW by 2050, would equate to c10PWh of additional demand (c23PWh gross minus c13PWh from higher efficiencies) and 2050e demand of c79PWh, 14% higher.

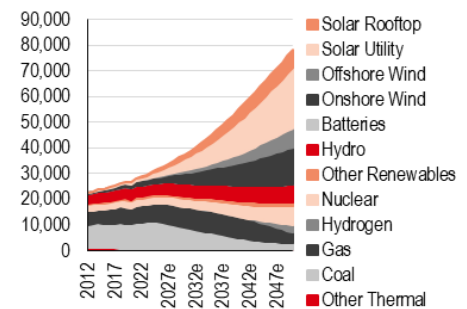
Current: Demand at c69PWh in 2050

HSBC global Load (TWh)



Blue-sky: 2050 demand 14% higher

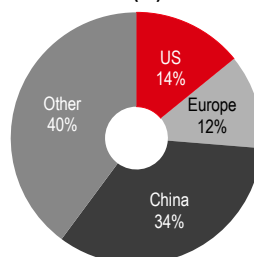
HSBC global Load (TWh)



In building our expectations of terrestrial power demand it is important to highlight our assumptions that AIDC demand is heavily skewed to the US (40%), Europe (30%) and China (20%), which is very different from today's power mix, where US and Europe as service/knowledge economies with a low electricity demand to GDP ratio only account for 12-14% of global demand, respectively.

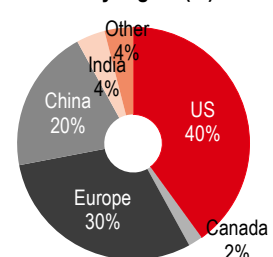
Together, US/Europe account for c26% of power demand today...

2026 Power demand split by region (%)



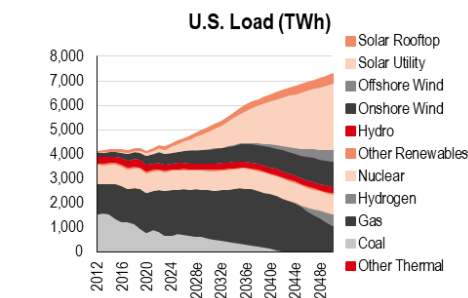
...but could capture 70% of AIDC growth, assuming AIDC demand is skewed to US, EU, other developed markets and China

Next 25 year power demand from AIDC by region (%)

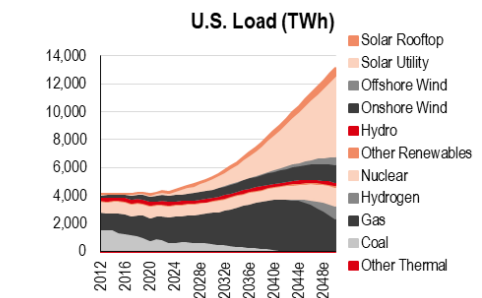


The implication would be an unprecedented power boom market in the US and Europe specifically where our annual growth expectation for the next 25 years would move from 2-3% to 4-5% and essentially match growth outlooks of emerging markets. Instead of rising by 60%, US power demand by 2050 would rise by 2.8x. European demand would also rise by 2.8x by 2050 (up from expectations of an 80% rise). Chinese demand would increase by 2.4x instead of 2.3x by 2050.

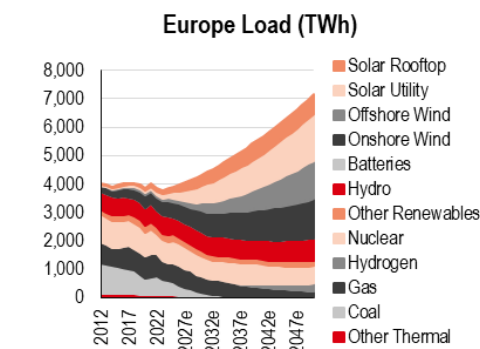
Current: +60% by 2050e



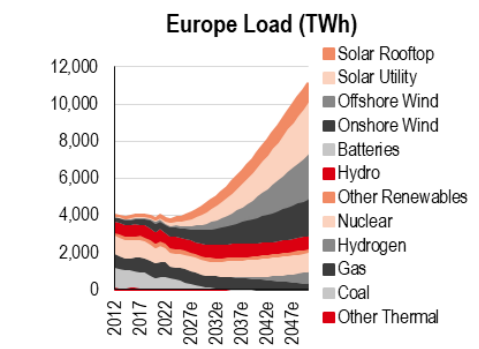
Blue-sky: 2.8x by 2050e



Current: +80% by 2050e



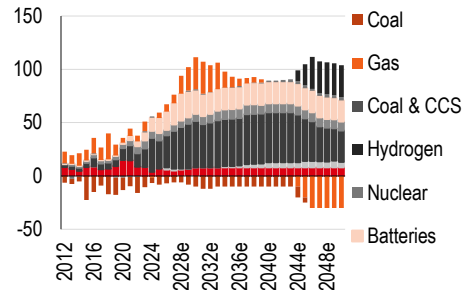
Blue-sky: 2.8x by 2050



Instead of US capacity additions topping out at around 100GW per annum with a 10-year spurt in gas (eventually starting to be replaced by hydrogen) and moderate net nuclear additions, US additions continue rising to c300GW per annum: Renewables capacity additions are 2-3x higher than before, gas additions extend longer, but in the 2040s are replaced by hydrogen and nuclear additions. The use of gas for growth in the medium term implies that carbon emissions stay at current levels in the US power system and continue to fall only when nuclear and hydrogen come into play by 2040.

Current: Additions rise to c100GW

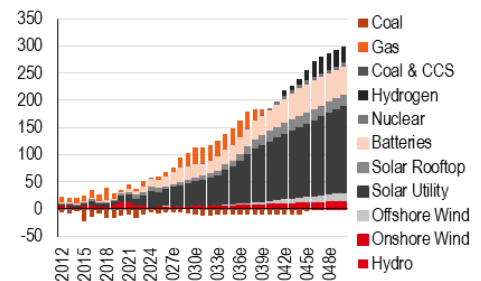
U.S. Capacity Changes (GW)



Source: Ember, HSBC estimate

Blue-sky: Additions rise to c300GW

U.S. Capacity Changes (GW)



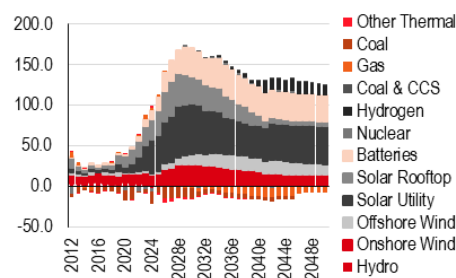
Source: Ember, HSBC estimate

We expect European capacity additions to rise to 170GW by 2030, before falling to 120GW. Yet, in a blue-sky scenario, additions rise to 280GW by 2050

On our current growth assumptions, European capacity additions rise to 170GW by 2030 on decarbonisation and growth and fall to 120GW per annum by 2040 primarily relying on growth as decarbonisation has largely been achieved. In the blue-sky view, on double the growth rate (especially from 2035 onwards), additions keep rising to c250GW per annum rather than falling. We model nuclear capacity of 149GW by 2050 rather than falling to c90GW (versus 115GW today), mainly based on extending the life of existing assets plus 16GW of new French capacity (10GW planned for 2038-45 plus 6GW additional in 2045-50e). We double to triple our renewables additions expectations and continue to model hydrogen by 2040+. Incremental growth via zero carbon technologies keeps the power carbon reduction trajectory on target. Compared to the US, we model a higher share of wind and particularly offshore wind and no additional new gas beyond already existing plans.

Current: Additions rise to c170GW and then fall to 120GW

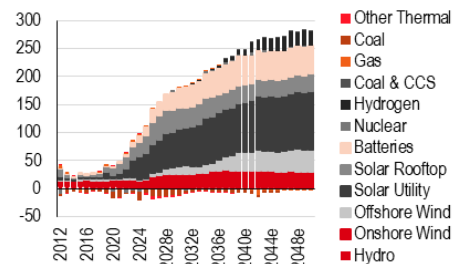
Europe Capacity Changes GW)



Source: Ember, HSBC estimate

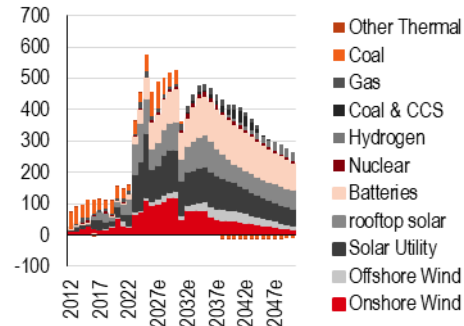
Blue-sky: Additions rise to c280GW

Europe Capacity Changes GW)

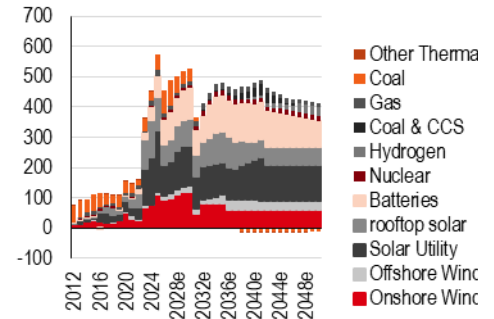


Source: Ember, HSBC estimate

With capacity additions of over 500GW in 2025, China has already scaled up additions to meet what is needed for substantial growth and decarbonisation. Under our current growth assumptions, 2025 is the peak year of additions, with a long-term declining trajectory. In the blue-sky scenario, we model that the uprate in Chinese demand mainly benefits renewables and nuclear. Higher growth long-term leads to steady rather than falling additions.

Current: Additions fall beyond 2030
China capacity changes (GW)


Source: Ember, HSBC estimate

Blue-sky: Additions remain steady
China capacity changes (GW)


Source: Ember, HSBC estimate

Arguments in favour of high terrestrial growth

AIDC demand is facing a power sector that is primed for growth. Storage is at the cusp of a boom and solar and onshore wind supply chains have overcapacity. The offshore wind market is maturing. While it takes a little while to re-accelerate gas and nuclear in the US/Europe (and requires some government direction for nuclear), gas and nuclear can contribute to growth. Networks need to be built as well, which might require some ramping in the US, yet this can be addressed by 2030/35. There are no strategic bottlenecks; no new technologies that need to be developed; it is rather a scaling of established businesses. We see Small Modular Reactors (SMRs) as an evolution, i.e., downscaling for industrial production of known large-scale nuclear technologies. SMRs are also rather an option and a failure to achieve them on time and at competitive economics doesn't change the sector's ability to support AIDC growth. Yet it is an important point to note: Power technology, nuclear/SMR and otherwise, is not static and there will continue to be technology progress across all segments – from solar and storage to wind and gas – which would support higher growth.

With or without additional long-term terrestrial AIDC demand, on our numbers China has already reached a new plateau of additions and companies are looking for export opportunities, which could support continued ramping of US/EU growth.

If India and China can do it, why not the US/EU? While 4-5% CAGR for power demand is unheard of in Europe and US, more than that has been business as usual for India or China in the past decade. European and US capacity supply chains and additions are already booming on decarbonisation alone and we see lack of growth as a downside risk to additions post 2030. China has scaled power additions from less than 100GW per annum in 2010-2020 to over 500GW per annum in 2025. Why wouldn't the US and Europe be able to continue to scale additions to 250-300GW pa by 2050 when additions have already scaled from levels of 30-40GW per annum 10 years ago to 50-100GW pa today with expectations of reaching 150-170GW pa by 2030?

Critical considerations

While it is technically possible, the speed of the build-out in some segments such as networks or nuclear might still be structurally too slow.

Growth for nuclear, offshore wind and a re-acceleration for onshore wind might require more supportive US energy policy and in the case of nuclear potentially guarantees and state investments – consistently so for decades.

Recent nuclear projects in the US/Europe have a track record of cost and timing overruns. There is general construction risk.

Power perspective – orbital could be a sensible hedge against policy and execution risk

While a marked scale up of power capacity additions is technically possible, there are policy and execution risks including timeliness. Orbital AIDC might be a good hedge.

Orbital AIDC benefits the solar supply chain

Companies and investors maintain a cautious view on orbital AIDC. Yet, on 1.2x-3.5x solar per IT load and 3x solar cost vs Earth, even moderate volumes could be an interesting market

In various reports we deep dive into how space solar would power and benefit from orbital AIDC and comment on Tesla's 100GW per year of solar manufacturing capacity goal in three years.

The following reports summarize feedback and views from different companies during SNEC (Photovoltaic Power Conference & Exhibition in Shanghai).

- ◆ **Space solar:** Our Asia Energy Transition team met with various space-level perovskite cell products from different companies during SNEC and believes adoption in space could be difficult and slow given concerns over reliability and the time needed to gather enough data from space to perfect the product. As a result, they believe the adoption of perovskite cells here on Earth will be quicker. Compared with silicon-based solar cells, perovskite cells are lighter and can generate 10-18% more power in weak-light environments. These features make them ideal for applications such as building integrated PVs (BIPVs), power for smart door locks, and portable electronic devices. For more details see [China Renewables SNEC takeaways: ESS strong, solar weak](#), 9 June 2026.
- ◆ **Conservative around SpaceX's 100GW per year of solar capacity goal:** Following an earlier report on space solar, our Asia Energy Transition team found that messaging from the SNEC tour implied conservative attitudes from both companies and investors regarding SpaceX and Tesla's 100GW per year of solar manufacturing capacity goal in three years in the US. For more details see [China Solar: Takeaways from SNEC 2026](#), 9 June 2026.

Previous reports deep dive into technology aspects, ambitions, upside to the solar supply chain and milestones. Experts from industry thought that SpaceX's AI Sat Mini may require up to 3.5x solar power over compute power load while public information related to the Starlink satellite AI1 model suggests a 175kw IT load with 210kw solar power system and potentially storage. As solar cells in space would likely still be 3x more expensive than the solar module on the ground in 15 years, space-based solar could be an interesting market. For more details please see [Space-based Solar: Read-across from SpaceX's latest dynamics](#) (1 June 2026) and [China Aerospace: Rocketing high](#) (12 February 2026).

Nemesis or best friend? The gas and nuclear conundrum

Hype cycles

The power sector is familiar with hype cycles. It can be differing technologies and has been coal, gas, nuclear, renewables and hydrogen in the past. It typically follows the same pattern of an initial focus on the total addressable market (as we are doing now) with estimates testing the upper bounds of what could be possible. Supply chains do ramp up (to some extent) on positive outlooks to be disappointed later as demand is below maximalist expectation and overheating competition does the rest – this holds for technology competition as well as on the company level.

The power sector is familiar with hype cycles, and it typically starts with initial focus on total addressable market (as we are doing now)

In situations of overcapacity after a potential hype cycle, supply chains may suffer more than utilities (if less diversified). Technologies with longer lead-times such as nuclear (less likely to scale up) or fuel cost such as gas (dispatched last) carry more risk. The good news is that sharing of risk and contracting is markedly different today. Today, there is a very high share of long-term contracts and thus offtakers carrying a higher share of risk versus earlier hype cycles, where utilities took on the risk of overcapacity via exposure to power markets and political risk.

If orbital AIDC gets built for strategic reasons such as a stepping stone for further space ambitions, i.e., scale-up is not only driven by economics, it could contribute to oversupply and thus margin pressure.

Trade-offs in favour of other regions

We note that we find it easier to model growth in the European power system (compared to the US) given the visible and concerted effort of network investments as well as consistent energy policy. Policy towards data centres is largely positive with governmental growth ambitions and utilities (and governments) vying for AIDC business. Thus, our modelling implies some spill-over of US AIDC to Europe on easier grid connections and greener power.

Another trade-off could imply more AIDC outside of US/EU/China. Would we go to orbit before building data centres in other regions of the world?

Trade-offs in favour of orbital

We think renewables would benefit from higher AIDC demand no matter what and solar most – terrestrial and orbital. However, as SpaceX's orbital AIDC business evolves, could this pit Elon Musk's interests against nuclear, SMRs and gas?

- ◆ The more growth we put in our power model, keeping gas output higher for longer or even leading to new gas generation, the further we move away from achieving emission reduction in the power sector, particularly in the US. Considering emissions, would orbital AIDC beat new gas additions on Earth?
- ◆ Might terrestrial nuclear in the US/Europe lose the race to orbital AIDC? Nuclear is zero carbon, but faces other challenges such as waste management, high cost in the US/EU (of up to EUR10-15bn/GW) and by far the longest ramp-up time of all technologies. In Europe and the US, 2035 is the timeline for early stage SMR prototypes and not quite yet for industrial production which is more likely by 2040+, if there ever is a business case. In 2026, France and EDF announced plans for c10GW of new nuclear capacity which would be available at the absolute earliest by 2037-41e. Yet, only a few months after the announcement, the timeline seems to be slipping already on lengthy approvals, and we model these 10GW by 2041-45. In parallel, EDF would also manage a large life extension program for the existing fleet. Hence, we rather see risks of timelines slipping, even beyond five years (as has been the case in EDF's recent projects), as opposed to an ability to accelerate. If SpaceX timelines hold, orbital AIDC might beat US/EU nuclear/SMR on timeliness.

In an alternative blue-sky scenario, we could also envision terrestrial AIDC scaling up to 100GW of additions and orbital AIDC reaching 50GW of additions per annum. In this scenario, we would model higher orbital AIDC to the benefit of space solar and at the expense of nuclear ramping in the 2040s and long-term gas/hydrogen output, before cutting renewables additions.

If SpaceX timelines hold, orbital AIDC might beat US/EU nuclear/SMR on timeliness

Disclosure appendix

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1. This report was produced and signed off by the author on 07 Sep 2026 11:38 GMT.
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